

Thermodynamic identifiability and unstable entropy inference under partial observation

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(Dated: 15 September 2026)

Exact identifiability of partially observed Markov dynamics does not by itself guarantee stable inference or a finite upper bound on microscopic entropy production. For finite irreducible continuous-time Markov chains with one resolved reverse pair, two distinct positive Laplace arguments identify entropy throughout the class with at most three states and recover the generator at every complete triangle. Three are sufficient for generator recovery throughout that class and necessary in the uncalibrated worst case. We distinguish this exact inverse from relative-rate conditioning and finite-trajectory estimation. Near a sparse boundary, generators and their observed laws converge while the stationary network entropy-production rate diverges, even with fixed trace and a common upper rate cap. More generally, at fixed state count and unknown reciprocal support, a finite local entropy ceiling exists precisely for two states or a complete three-state source. We derive an exponential observation-time lower bound for testing independently variable rare reverse rates and implement a constrained finite-data inference pipeline. For a five-state family we exhaust all admissible supports at and above a threshold, obtaining exactly two entropy values and then a unique generator; below it we prove an attainable entropy half-line, without claiming the complete entropy set. A nontrivial perturbation clarifies its boundary-feasibility transition without asserting a generic saddle-node of the whole realization set. A finite state cap and closed positive supported-rate bounds give attained upper entropy bounds converging to their exact-data value as tolerance shrinks. All exact results retain explicit dimension and support constraints; numerical estimation and continuation are distinguished from proof.

I. INTRODUCTION

Partial observations can reveal broken time-reversal symmetry without resolving every dissipative transition. Waiting times between visible jumps retain information about hidden excursions and can improve lower bounds on entropy production [1–3]. We ask the complementary question: can these observations exclude arbitrarily large microscopic entropy production? An upper bound matters whenever a low observed irreversibility, or a well-fitting reconstructed model, is used to argue that little dissipation remains hidden.

Five questions must be separated. Do exact observations determine the generator? Do they

determine its entropy production even if the generator is ambiguous? Is the compatible entropy set bounded? Does a small observational error give a small change, or at least a finite ceiling? Finally, can a finite trajectory support useful estimates and uncertainty intervals? An affirmative answer to the first question need not answer the last three. The rate-to-entropy map can itself become singular as a reverse flux vanishes, while the observation inverse introduces additional ambiguity.

Our central result is a sharp local upper-ceiling criterion for one resolved microscopic reverse pair, fixed state count, and unknown reciprocal support. A finite ceiling in a neighborhood of the joint waiting laws exists exactly for two states or a complete three-state source. The criterion remains true with observed rates and trace fixed. This general statement organizes the paper. A three-Laplace inverse without a rate cap provides the positive three-state result; a sparse family separates stable absolute-rate recovery from unstable entropy. Its rare-rate alternatives also yield an exponential lower bound on the duration needed to distinguish a fixed multiplicative entropy gap. We complement these results with numerical conditioning and finite-trajectory inference, retaining their statistical and model-class limitations.

The physical and mathematical antecedents are substantial. Data processing bounds observed path irreversibility by microscopic entropy production [1, 2]; hidden nonequilibrium and compatible-model minimization already underlie partial-observation lower bounds [4–6]. Similarity descriptions of hidden Markov and Markov arrival models, phase-type invariant polytopes, and finite nonnegative factorization ambiguity provide established realization tools [7–10]. Neither positive-realization ambiguity nor relative-entropy singularity is new [11, 12]. Our contribution concerns a support-exhaustive two-value example and precise local upper-inference obstructions in stated physical model classes. We do not claim a new general realization theory or a full description of every subcritical entropy set.

An explicit five-state family remains a second exact result, presented after the general stability and inference results. Below the threshold its entropy set is unbounded above; at the threshold there are exactly two generators with distinct entropy values, and above it the generator is unique. The proof exhausts reciprocal physical supports where a finite count is claimed; finite numerical search does not establish that count. A two-parameter extension tests how the threshold persists and distinguishes the fold of boundary constraints from the continuum of interior realizations. Adding one nonminimal hidden state destroys the finite fiber. This dimensional dependence is part of the conclusion, not an auxiliary qualification. The appendices preserve the full global proof and exact certificates; the expanded source comparison and numerical methods document what is and is not established.

A. Relation to inference and realization literature

Much thermodynamic inference asks for a lower bound: the least microscopic dissipation compatible with observed currents, coarse transitions, or waiting times. Thermodynamic uncertainty relations constrain this least cost from current fluctuations [13, 14]; semi-Markov and waiting-time methods use further temporal information [1, 2, 15]. Compatible-model minimization makes the lower-bound problem explicit [5, 6, 16, 17]. Ehrich’s masked discrete-time example already reports apparently unbounded compatible entropy as a reverse transition probability approaches zero [5, Sec. 4.1, Fig. 8]. Nitzan et al. explicitly study constraint tolerance in hidden-model fitting [6, Sec. III.C]. Neither compatible-model optimization nor the possibility of hidden entropy divergence is claimed as new here. Our question concerns the upper end of the same type of inverse problem: can observations exclude arbitrarily large microscopic entropy production? A valid lower bound, even one that is sharp under a chosen topology, does not answer that question over a larger model class. Conversely, one observed reverse pair can already recover total entropy when the network is known to be unicyclic [1]. That topology-restricted statement does not classify compatible generators over unknown reciprocal supports.

Maier, Seifert and van der Meer bound entropy along individual hidden paths and reconstruct candidate minimal graphs from transition waiting times [3, Secs. III, VI–VII]. Their minimal graph combines shortest-path information within the physical network; minimal linear realization order instead measures the dimension needed to represent the entire transfer function. These are different notions of minimality. Our identified set ranges over all compatible generators in a declared state class and concerns their stationary entropy-production rates.

Finite observation is a separate limitation. Recent work compares waiting-time and current estimators under finite statistics, blurred labels and temporal binning [18]. Spatial lumping and blurred transitions reveal additional resolution-dependent dissipative scales [19]. In the temporal-binning model of Fritz et al., every event and its order remain recorded; finite-sample entropy estimates can overestimate despite the corresponding population lower bound. We do not identify such binning with lost events or use their error analysis as coverage for our sequential rate estimator. We instead analyze reconstruction of microscopic rates and the resulting entropy uncertainty within a specified identifiable class. The distinction between structural identification and statistical learnability is already well developed for hidden Markov models [20, 21]. Our sparse-boundary example isolates another obstruction: the rate inverse can remain continuous while the entropy functional becomes singular.

The observation model matters mathematically and physically. Aggregated Markov models admit equivalent canonical representations that need not preserve nonnegative microscopic rates [22]; positive HMM, MAP and phase-type realization theory provides related similarity and cone methods [7–9, 23]. General coarse trajectories need not have our resolved-edge reset property. Their physical time reverse also requires care when the coarse-graining is nonlocal in time [24, 25]. Recent continuous-space estimators address observations without microscopic reset events [26]; those experiments are outside the finite-state inverse problem considered here.

Hidden entropy and its fluctuation relations also depend on the microscopic time-reversal convention [27]; they do not provide an upper ceiling from our partial waiting laws.

II. MODEL, OBSERVATIONS, AND IDENTIFIED SETS

Fix a physical time unit and two named states x, y . For $N \geq 2$, let $\mathcal{C}_N$ consist of row generators Q on $\{x, y, h_1, \dots, h_{N-2}\}$ with $Q\mathbf{1} = 0$, nonnegative off-diagonal entries, connected reciprocal support

$$q_{ij} > 0 \iff q_{ji} > 0, \quad i \neq j, \quad (1)$$

and $r = q_{xy} > 0$, $s = q_{yx} > 0$. Every rate is finite. There is no common rate floor or upper bound unless stated, and competing support graphs are unknown. Set $\mathcal{C}_{\leq m} = \bigcup_{N=2}^m \mathcal{C}_N$. A fixed count and a cap are different assumptions.

There is one kinetic channel per ordered state pair, the generator is time independent, and all states are even under time reversal. The generator entropy-production functional is

$$\sigma(Q) = \sum_{i < j: q_{ij} > 0} (F_{ij} - F_{ji}) \log \frac{F_{ij}}{F_{ji}}, \quad F_{ij} = \pi_i q_{ij}, \quad (2)$$

$$= \sum_{i \neq j: q_{ij} > 0} \pi_i q_{ij} \log \frac{q_{ij}}{q_{ji}}, \quad (3)$$

where $\pi Q = 0$, $\pi\mathbf{1} = 1$ is the unique positive stationary distribution. Equation (3) follows by stationary cancellation of the $\log \pi_i$ terms. We use natural logarithms and $k_B = 1$, so σ has inverse-time units. A paired absent edge contributes zero. Every member of $\mathcal{C}_N$ has finite nonnegative entropy, zero exactly at detailed balance. This is the usual finite-network convention [28, 29]; interpreting it as all physical dissipation additionally requires resolved thermodynamic channels and local detailed balance. Parallel unresolved channels or odd variables require a different model.

Observe only marks $+$ $= (x, y)$ and $- = (y, x)$, with known endpoints. Define

$$E_+ = re_x e_y^T, \quad E_- = se_y e_x^T, \quad T = Q - E_+ - E_-, \quad (4)$$

$$R = \begin{pmatrix} e_y^T \\ e_x^T \end{pmatrix}, \quad B = (re_x, se_y). \quad (5)$$

Crucially, T retains the diagonals of Q . Killing at a marked jump makes T transient. The observation is the matrix of joint next-mark/time densities

$$\mathcal{O}(Q) = \Psi_Q(t) = Re^{Tt}B, \quad \widehat{\Psi}_Q(\lambda) = R(\lambda I - T)^{-1}B, \quad \lambda > 0. \quad (6)$$

Each row is a probability measure $\mu_I(dt, J) = \psi_{IJ}(t)dt$ on $[0, \infty) \times \{+, -\}$. Individual entries need not integrate to one. Absolute time and joint amplitudes are part of the data. The reset property determines all successive marked-event laws from these rows [1, 2].

Two models are observationally equivalent when their two row laws agree, equivalently when $\Psi_Q(t) = \Psi_{Q'}(t)$ for every $t \geq 0$. Physical models are identified only under permutations of hidden states fixing x, y . For a declared class $\mathcal{C}$, define

$$\mathcal{F}_\mathcal{C}(D) = \{[Q]_{\text{lab}} : Q \in \mathcal{C}, \mathcal{O}(Q) = D\}, \quad \mathcal{S}_\mathcal{C}(D) = \{\sigma(Q) : [Q]_{\text{lab}} \in \mathcal{F}_\mathcal{C}(D)\}. \quad (7)$$

Generator identification means $|\mathcal{F}_\mathcal{C}(D)| = 1$; thermodynamic identification means $|\mathcal{S}_\mathcal{C}(D)| = 1$. A bounded nonunique entropy fiber has more than one value and finite supremum. An unbounded fiber contains arbitrarily large *finite* values; no model with a one-way edge is admitted. Unboundedness alone neither gives its infimum nor specifies every attainable value.

For approximate data we use maximum row total variation,

$$d(D, D') = \max_I \frac{1}{2} \sum_J \int_0^\infty |\psi_{IJ}(t) - \psi'_{IJ}(t)| dt. \quad (8)$$

Equation (8) compares underlying laws. A finite unsmoothed empirical waiting law is atomic, whereas every row law here has a time density, so their TV distance is exactly one. A small TV ball centered on that empirical law is therefore not an operational confidence region. Section VIB instead uses bounded Laplace statistics and states their own sampling guarantees.

We also consider the product weak topology of the two row probability measures. A *finite local upper ceiling* at D in $\mathcal{C}$ means that some $\eta > 0$ satisfies

$$\sup\{\sigma(P) : P \in \mathcal{C}, d(\mathcal{O}(P), D) < \eta\} < \infty. \quad (9)$$

Its weak version replaces the ball by an open neighborhood. This property is defined even when exact entropy is multivalued. At an identifiable source, convergence of entropy along all observationally convergent models is stronger than a finite ceiling. Stationary records on a fixed time window form a separate experiment, treated in Sec. VI.

TABLE I. Properties of an inverse problem for a declared model class. Conditioning requires a parameterization and data scale; finite-sample inference additionally requires a sampling experiment.

Property	Requirement
Generator identification	The generator fiber has one element
Entropy identification	The entropy fiber has one value
Exact boundedness	The exact entropy fiber has finite supremum
Local upper ceiling	A positive data neighborhood has finite entropy supremum
Continuity	Entropy converges along admissible converging data
Conditioning	Quantitative sensitivity to perturbations in specified coordinates
Finite-sample estimation	Error and coverage controlled under a sampling model

The dimension assumption must accompany every identified set and must come from external physical information or an explicit modeling restriction; it cannot be inferred as an upper bound from the observable realization order. A fixed-dimensional class, a state cap, and a class of linearly minimal realizations are different restrictions. The smallest positive realization can in general require more states than the minimal linear order. For our particular five-state family those orders coincide, so its cap may be replaced by linear minimality or by requiring a minimum-size positive realization. This equality is proved for that family, not assumed for arbitrary kernels. With unrestricted hidden dimension a fully hidden state can be split into copies, preserving the complete marked law while generating unbounded entropy (Appendix F).

At stationarity Eq. (2) is both the pathwise Markov irreversibility rate and the mean total network entropy-production rate. The stationary system-entropy derivative vanishes, so the corresponding medium-entropy rate agrees with Eq. (3). These are mathematical identities of the specified CTMC. Interpreting each rate ratio as reservoir entropy requires local detailed balance and correct microscopic state and channel resolution; a sum of parallel channels generally loses that information. The generator alone need not specify unique heat currents or heat dissipation in physical energy units.

III. EXACT THREE-STATE RECOVERY AND CONDITIONING

Three states isolate the difference between recovering rates and recovering entropy. There is only one hidden state, so no nontrivial hidden similarity remains, but this algebraic observation alone would not establish stability.

Theorem 1 (Three positive Laplace arguments). *For every $Q_0 \in \mathcal{C}_3$ and any $0 < \lambda_1 < \lambda_2 < \lambda_3$, the three matrices $\hat{\Psi}_{Q_0}(\lambda_j)$ determine Q_0 uniquely within $\mathcal{C}_{\leq 3}$. On admissible data near these matrices, the inverse is rational with nonzero denominators and locally Lipschitz. Consequently weak convergence of joint row laws to $\mathcal{O}(Q_0)$ among models in $\mathcal{C}_{\leq 3}$ implies eventual dimension three and entrywise convergence to Q_0 . In row TV the inverse is locally Lipschitz. At a complete three-state source, entropy is also locally Lipschitz in row TV and continuous under weak row convergence.*

Proof. Write

$$Q = \begin{pmatrix} -r-a & r & a \\ s & -s-c & c \\ b & d & -b-d \end{pmatrix}, \quad h = b + d > 0. \quad (10)$$

Reciprocity and irreducibility imply $a + c > 0$. Let P interchange the two mark rows, $F(\lambda) = P\hat{\Psi}(\lambda)$, and $M(\lambda) = F(\lambda)^{-1}$. The visible resolvent K satisfies $F = K \text{diag}(r, s)$ and

$$K^{-1} = \text{diag}(\lambda + r + a, \lambda + s + c) - \frac{1}{\lambda + h} \begin{pmatrix} a \\ c \end{pmatrix} (b, d). \quad (11)$$

This is a nonsingular strictly row diagonally dominant matrix for $\lambda > 0$. Hence

$$G_x(\lambda) := \frac{(M\mathbf{1})_x - 1}{\lambda} = \frac{1}{r} + \frac{a/r}{\lambda + h}, \quad G_y(\lambda) = \frac{1}{s} + \frac{c/s}{\lambda + h}. \quad (12)$$

Select a row with positive *source* entrance and write it as $G = u + v/(\lambda + h)$, $v > 0$. If $G_j = G(\lambda_j)$, then

$$\mathcal{R} = \frac{G_1 - G_2}{G_2 - G_3} \frac{\lambda_3 - \lambda_2}{\lambda_2 - \lambda_1} = \frac{\lambda_3 + h}{\lambda_1 + h} > 1, \quad h = \frac{\lambda_3 - \mathcal{R}\lambda_1}{\mathcal{R} - 1}. \quad (13)$$

For each row set

$$v_i = (G_i(\lambda_1) - G_i(\lambda_2)) \frac{(\lambda_1 + h)(\lambda_2 + h)}{\lambda_2 - \lambda_1}, \quad u_i = G_i(\lambda_1) - \frac{v_i}{\lambda_1 + h}. \quad (14)$$

Then $r = 1/u_x$, $s = 1/u_y$, $a = v_x/u_x$, and $c = v_y/u_y$. If $a_0 > 0$, use

$$d = -rM_{xy}(\lambda_1)(\lambda_1 + h)/a, \quad b = h - d. \quad (15)$$

Otherwise $c_0 > 0$ and use $b = -sM_{yx}(\lambda_1)(\lambda_1 + h)/c$, $d = h - b$. Keep the source-selected branch throughout its neighborhood. No formula divides by an entrance vanishing at a leaf.

All denominators are nonzero at the source and remain so locally. The inverse is therefore locally Lipschitz as a function of the finitely many transform entries. Those entries are bounded continuous tests of row laws and are Lipschitz in TV. Two-state data have both G rows constant and cannot approach the selected nonzero source differences. At a complete source the recovered rates stay bounded away from zero, and the normalized stationary solution and (2) are smooth. This proves the entropy statements as well. $\square$

No trace or rate cap is assumed in this reconstruction. It does not require differentiation of noisy waiting-time laws. The following refinement separates two-point entropy identification from generator recovery.

A. Two-point entropy identification and generator ambiguity

Proposition 2 (Two arguments for complete triangles). *For every complete $Q \in \mathcal{C}_3$ and any $0 < \lambda_1 < \lambda_2$, the two full matrices $\hat{\Psi}_Q(\lambda_1), \hat{\Psi}_Q(\lambda_2)$ determine all six rates uniquely within $\mathcal{C}_{\leq 3}$, without separately calibrating r, s . The inverse is rational and locally Lipschitz at each such source.*

Proof. Use $M = (P\hat{\Psi})^{-1}$ and $h = b + d$ from the preceding reconstruction. The Schur complement gives

$$M_{xy}(\lambda) = -\frac{ad}{r(\lambda + h)}, \quad M_{yx}(\lambda) = -\frac{cb}{s(\lambda + h)}. \quad (16)$$

Both are nonzero at a complete triangle. Thus

$$\rho = \frac{M_{xy}(\lambda_1)}{M_{xy}(\lambda_2)} = \frac{\lambda_2 + h}{\lambda_1 + h} > 1, \quad h = \frac{\lambda_2 - \rho\lambda_1}{\rho - 1}. \quad (17)$$

The two-point formulas for u_i, v_i in the preceding proof now recover r, s, a, c , and Eq. (15) recovers b, d . All their denominators are nonzero at the source. A two-state competitor has zero inverse off-diagonals; a reciprocal three-state competitor with equal data must have $ad > 0$, which forces its complete support. It is therefore recovered by the same rational formulas, proving uniqueness in the full stated cap. $\square$

Corollary 3 (Two arguments identify entropy). *For every $Q \in \mathcal{C}_{\leq 3}$ and any $0 < \lambda_1 < \lambda_2$, the two full Laplace matrices determine $\sigma(Q)$ uniquely within $\mathcal{C}_{\leq 3}$, without separately calibrating the observed rates.*

Proof. A nonzero off-diagonal of $M = (P\hat{\Psi})^{-1}$ forces complete three-state support, so Proposition 2 recovers Q . Otherwise every compatible model is a reciprocal tree or two-state chain. Stationarity across each edge cut gives zero current, hence $\sigma = 0$. $\square$

The zero pattern refers to M , equivalently the off-diagonals of $F = P\hat{\Psi}$, not the off-diagonals of $\hat{\Psi}$. Exact entropy identification does not imply continuity at tree boundaries.

Three points remain necessary in the worst case for *generator recovery* with uncalibrated rates. At a hidden leaf attached to x , $r, s, a, b > 0$ and $c = d = 0$, so both off-diagonals in Eq. (16) vanish. The remaining tests are $G_x(\lambda) = u + v/(\lambda + b)$, with $u = 1/r$, $v = a/r > 0$, and $G_y = 1/s$. Given two distinct positive arguments, put

$$D = \frac{G_x(\lambda_1) - G_x(\lambda_2)}{\lambda_2 - \lambda_1} > 0, \quad v_\beta = D(\lambda_1 + \beta)(\lambda_2 + \beta), \quad (18)$$

$$u_\beta = G_x(\lambda_1) - D(\lambda_2 + \beta). \quad (19)$$

Every $\beta > 0$ with $u_\beta > 0$ yields a leaf generator with

$$(r, s, a, b, c, d) = (u_\beta^{-1}, s, v_\beta/u_\beta, \beta, 0, 0) \quad (20)$$

having the same two full matrices. An open interval around the source $\beta = b$ is admissible, and $u'_\beta = -D$ makes the generators distinct. The y -leaf is symmetric. Together with Theorem 1, this establishes a worst-case minimum of three positive arguments for generator recovery. In the present reciprocal three-state model with the observed x - y pair present, the two leaf attachments are the only noncomplete irreducible supports; the conclusion is not a classification of arbitrary sparse larger networks.

For an explicit check, using the tuple order (r, s, a, b, c, d) , take

$$\theta_1 = (1, 1, 1, 1, 0, 0), \quad \theta_2 = (6/5, 1, 12/5, 2, 0, 0). \quad (21)$$

Direct evaluation of $R(\lambda I - T)^{-1}B$, including the two observed-rate amplitudes and the $(+, -)$ row order, gives

$$\hat{\Psi}_1(1) = \hat{\Psi}_2(1) = \begin{pmatrix} 0 & 1/2 \\ 2/5 & 0 \end{pmatrix}, \quad \hat{\Psi}_1(2) = \hat{\Psi}_2(2) = \begin{pmatrix} 0 & 1/3 \\ 3/11 & 0 \end{pmatrix}. \quad (22)$$

The complete transforms differ, since

$$[\hat{\Psi}_1 - \hat{\Psi}_2]_{-,+}(\lambda) = -\frac{\lambda(\lambda - 1)(\lambda - 2)}{(\lambda^2 + 3\lambda + 1)(5\lambda^2 + 28\lambda + 12)}. \quad (23)$$

At $\lambda = 3$ their lower-left entries are $4/19$ and $10/47$. These are reciprocal trees and both have zero entropy. The example is ambiguity of the generator from a finite set of summaries, not equality of the full waiting laws or a nontrivial entropy fiber.

Calibration changes this comparison. The observed rate r is 1 versus $6/5$, whereas $s = 1$ in both. Their stationary mark intensities in physical time are $1/3$ per mark versus $6/17$ per mark; their normalized mark proportions are nevertheless $(1/2, 1/2)$ in both. Their traces also differ. Thus fixed observed rates, a supplied absolute stationary event intensity, or a supplied trace exclude this particular pair. More generally, if both r, s are known, subtracting the appropriate constant from G_i lets the ratio of two nonzero residual tests recover h even at a leaf. Two positive arguments then suffice for every three-state source in that calibrated class.

The accompanying exact audit retains the one-point generator counterexample and Jacobian diagnostics. That equilibrium example does not establish that two arguments are necessary for entropy identification. The three-point stability theorem retains its separate role near leaf boundaries.

B. Conditioning of the three-state inverse

Local Lipschitz inversion is a pointwise statement. To quantify its constants, write $\theta = (r, s, a, c, b, d)$ and let $\phi(\theta) \in \mathbb{R}^{12}$ collect the joint transform entries at three distinct positive arguments. With $K_\lambda = (\lambda I - T)^{-1}$, its full six-rate derivative is

$$\partial_j \hat{\Psi}(\lambda) = RK_\lambda(\partial_j T)K_\lambda B + RK_\lambda(\partial_j B). \quad (24)$$

The second term is essential when the observed rates r, s are unknown. Absolute-rate errors are governed by $J_\theta = D\phi$, whereas relative-rate errors are governed by $J_{\log} = J_\theta \text{diag}(\theta)$. These describe different inverse questions. Raw singular values carry time units; logarithmic coordinates are dimensionless in the fixed time convention.

The rational inverse in Theorem 1 has a derivative that is a left inverse of J_θ . This remains true at an irreducible tree by continuity from the positive interior. Consequently the raw derivative has rank six there. Sparse limits need not make absolute-rate recovery ill conditioned. For example, at the unit tree and $\lambda = (0.2, 1, 5)$, its smallest singular value is approximately 0.00744.

For the family (27), fix $0 < e < 1/2$ and let $k \downarrow 0$. The rational formulas also remain nonsingular at this irreducible one-way limit, although the limiting model itself is outside our reciprocal class.

Thus J_θ converges to a full-rank matrix while

$$s_{\min}(J_{\log}) = \Theta(k), \quad \text{cond}(J_{\log}) = \Theta(k^{-1}). \quad (25)$$

Indeed $s_{\min}(J_\theta D) \geq s_{\min}(J_\theta) \min_j D_{jj}$ for $D = \text{diag}(\theta)$, and testing the unit b coordinate gives the matching upper bound. The other five rate coordinates stay positive, so the limiting nullspace of $J_{\log}$ is exactly the $b = k$ direction. Its weakest right singular vector converges to that direction, up to sign. Meanwhile $\partial_b \sigma = -J_0(e)/b + O(\log(1/b))$, with $J_0(e) = e(1 - e)/(3 + e - e^2) > 0$. Relative-rate and entropy sensitivity therefore diverge even where absolute-rate inversion remains regular.

The choice of Laplace arguments also matters. As three distinct arguments coalesce, the limiting repeated transform matrix has at most four independent entries for six rates, so a singular value tends to zero. Arguments all tending to zero or infinity similarly lose informative variation. Hence “any three distinct positive points” guarantees exact recoverability, with no uniform conditioning guarantee over designs or sources.

The scalar inverse already shows this source dependence. Differentiating Eq. (13), and taking differences in Eq. (12), gives

$$\frac{\partial h}{\partial \mathcal{R}} = -\frac{(\lambda_1 + h)^2}{\lambda_3 - \lambda_1}, \quad G(\lambda_i) - G(\lambda_j) = \frac{v(\lambda_j - \lambda_i)}{(\lambda_i + h)(\lambda_j + h)}. \quad (26)$$

Fast hidden escape relative to the sampled Laplace scales amplifies errors in $\mathcal{R}$. A small entrance amplitude v or large h also suppresses the differences used to form that ratio. Thus a finite local inverse constant need not give useful precision at an experimental noise level.

We compared all 165 triples from $\{0.025, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10, 20, 40\}$ at the reference in Fig. 1. The largest minimum log singular value occurs at $(0.2, 0.5, 5)$. The smallest oracle asymptotic entropy standard error for the moment estimator below instead occurs at $(0.025, 10, 20)$: its $\text{SE}\sqrt{n}$ is 1.176, compared with 1.238 for $(0.2, 1, 5)$. The better entropy design has a worse Euclidean condition number. These are finite-grid comparisons at one reference, not universal optima. A practical starting point is a separated logarithmic triple spanning a pilot waiting-time scale, followed by a covariance-aware assessment of the target uncertainty. Design selection from the same data needs its own selection correction or an independent pilot.

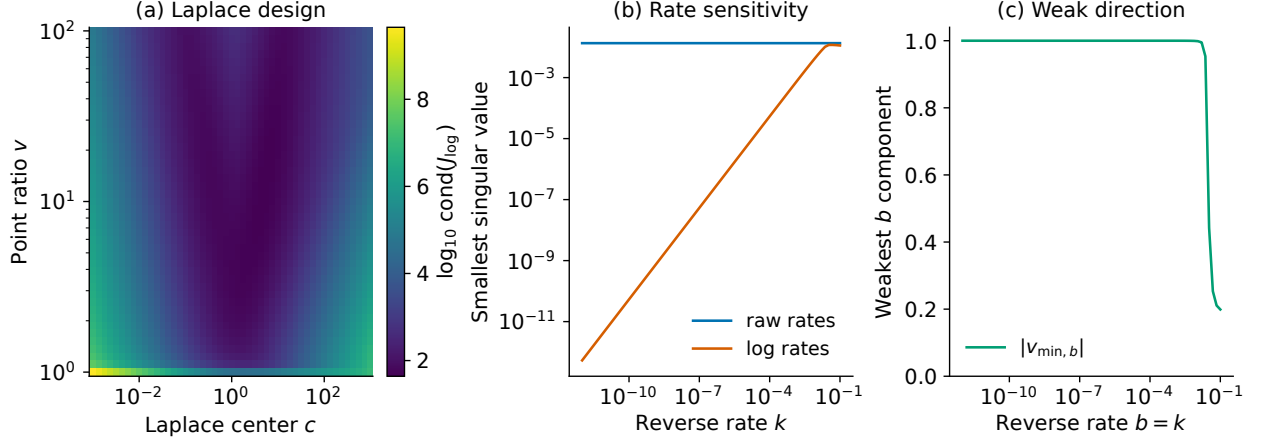


FIG. 1. Conditioning of all six unknown three-state rates. (a) Log-rate condition number for $(\lambda_1, \lambda_2, \lambda_3) = (c/v, c, cv)$ at $(r, s, a, c, b, d) = (1.2, 0.8, 0.9, 0.4, 0.3, 1.1)$; the horizontal c denotes the Laplace center, not the rate coordinate. Nearly coincident points and inappropriate scales degrade inversion. (b) At fixed $e = 0.2$ in Eq. (27), the raw minimum singular value stays positive as k decreases, whereas its log-rate counterpart scales as k . (c) The weakest log-rate direction tends to the reverse-rate coordinate $b = k$. Panels (b,c) use $\lambda = (0.2, 1, 5)$. These unweighted Euclidean diagnostics are not Fisher information; statistical noise must also be considered.

IV. ENTROPY INSTABILITY NEAR SPARSE BOUNDARIES

Proposition 4 (A fixed-trace three-state boundary). *For $0 < k < e < 1/2$, let*

$$Q(e, k) = \begin{pmatrix} -1 - e & 1 & e \\ 1 & -2 + k & 1 - k \\ k & 1 - e & -1 + e - k \end{pmatrix}. \quad (27)$$

These generators have trace -4 , off-diagonal rates at most one, and observed rates $r = s = 1$. Each exact kernel uniquely determines its generator in $\mathcal{C}_{\leq 3}$. With $Z = 3 + e + 2k - e^2 - ek - k^2$,

$$\sigma(Q(e, k)) = \frac{(e - k)(1 - e - k)}{Z} \log \frac{e(1 - e)}{k(1 - k)}. \quad (28)$$

Their row distance from the unit tree $Q(0, 0)$ is below $4e$. Along $k = \exp(-1/e^2)$, as $e \downarrow 0$,

$$Q(e, k) \longrightarrow Q(0, 0), \quad e\sigma(Q(e, k)) \longrightarrow \frac{1}{3}. \quad (29)$$

Every positive observation neighborhood of the tree has unbounded entropy even with the stated trace and rate cap.

Proof. The unnormalized stationary weights are

$$(w_x, w_y, w_h) = (1 - e + 2k - k^2, 1 + k - e^2, 1 - k + 2e - ek). \quad (30)$$

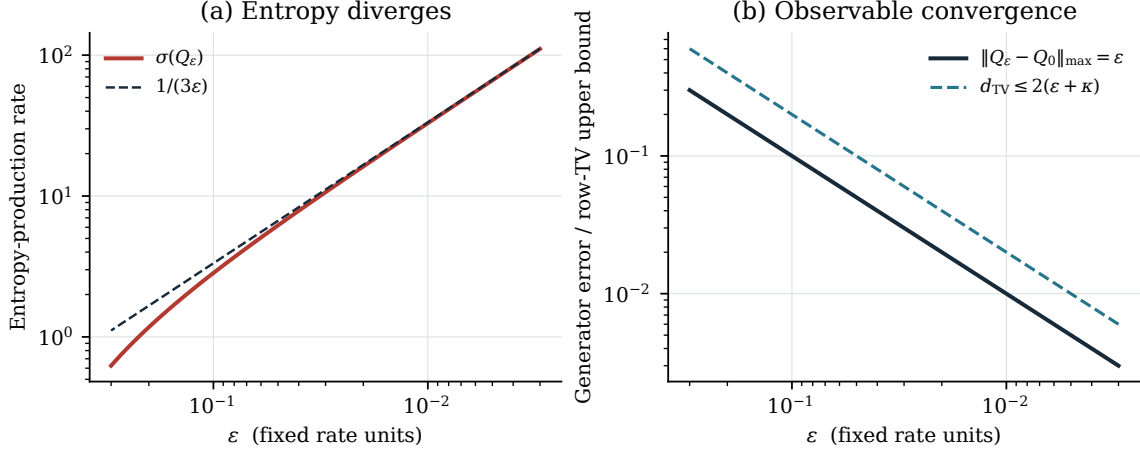


FIG. 2. The deterministic family (27) with $k = \exp(-1/e^2)$; the figure uses $\epsilon = e$ and $\kappa = k$. The rates converge to the unit tree and the analytic row-TV upper bound $2(e + k)$ tends to zero, while entropy diverges with asymptote $1/(3e)$. The plotted TV line is a bound, not a measured distance. All finite-parameter models retain strictly positive reverse rates, observed rates one, trace -4 , and off-diagonal rates at most one. Arbitrary-precision evaluation avoids replacing the small reverse rate by zero.

They sum to Z and solve $wQ = 0$. Along the cycle $x \rightarrow h \rightarrow y \rightarrow x$, each edge current is $(e - k)(1 - e - k)/Z$, and the cycle affinity is the logarithm in (28). This proves the entropy identity. The source's mean next-mark times from (x, y, h) are $(1, 2, 3)$. Common/excess-clock coupling has separation hazard at most $e + k$, hence row TV is at most $2(e + k) < 4e$. In (28), $Z \rightarrow 3$ and $\log(1/k) = 1/e^2$, giving (29). Alternatively, first fix any sufficiently small $e > 0$, then let $k \downarrow 0$: the logarithmic coefficient tends to $e(1 - e)/(3 + e - e^2) > 0$. This proves an infinite supremum in a fixed neighborhood. The two limiting statements are distinct. $\square$

In the sequence of Proposition 4, the common cycle current is $J = (e - k)(1 - e - k)/Z \sim e/3$, while its cycle affinity is $\mathcal{A} = \log[e(1 - e)/(k(1 - k))] \sim e^{-2}$. Thus $\sigma = J\mathcal{A} \sim 1/(3e)$ even though the current tends to zero and every microscopic rate is bounded. Bounded kinetic speed does not bound thermodynamic driving. This boundary singularity is already present when the full generator is observed with small entrywise error; partial observation is not needed for it. Partial observation adds a different effect: complete-support generators can have exactly the same waiting law while their entropies differ without bound. Appendix D gives an explicit four-state example and the general mechanism used by the subcritical five-state constructions. Every prelimit model is reciprocal; the limiting boundary supplies the entropy divergence.

More generally, at any noncomplete $Q_0 \in \mathcal{C}_N$, open a missing pair at rates e and $\exp(-1/e^2)$ and subtract their sum from an existing positive unobserved ordered rate. For small e this preserves

connected reciprocal support, observed rates, trace, and the original maximum rate. Stationarity varies continuously, so the new edge contributes asymptotically π_i^0/e . Thus generator-level entropy is locally upper bounded exactly at complete sources. A complete generator nevertheless need not give observational upper boundedness when more hidden states are present.

V. A SHARP CRITERION FOR FINITE LOCAL UPPER CEILINGS

For a source Q_0 , denote by $\mathcal{C}_N(Q_0; r, s, \tau)$ the subclass fixing its two observed rates and $\tau = -\text{tr } Q_0$. No further individual rate restriction is implicit.

Theorem 5 (Fixed-count local criterion). *For $Q_0 \in \mathcal{C}_N$ with one observed reverse pair and unknown competitor topology, a finite local entropy upper ceiling exists in $\mathcal{C}_N$ if and only if*

$$N = 2 \quad \text{or} \quad N = 3 \text{ and } Q_0 \text{ has complete support.} \quad (31)$$

The same criterion holds in $\mathcal{C}_N(Q_0; r, s, \tau)$ and in the product weak topology of joint row laws.

The large- N part follows from a more general constructive statement. For a finite set of resolved marks, extend (6) by $R_I = e_{\text{target}(I)}^T$ and $B_J = q_J e_{\text{source}(J)}$, with $T = Q - \sum_J E_J$. Extend (8) by maximizing over all previous marks and summing over all next marks.

Lemma 6 (Two fully hidden states). *Allow any nonempty reverse-closed set of individually resolved observed edges on a fixed state set. Suppose two distinct states are absent from every observed endpoint. For every source Q_0 , every $\eta > 0$, and every finite L , there is a complete bidirected generator Q' with the same state count, every observed rate, and trace, such that*

$$d(\mathcal{O}(Q'), \mathcal{O}(Q_0)) < \eta, \quad \sigma(Q') > L. \quad (32)$$

Every off-diagonal rate of the constructed models is at most $\tau = -\text{tr } Q_0$. No minimality assumption is needed.

Proof. Let h, k be fully hidden and $\mathcal{U}$ the unobserved ordered pairs. Their source rate mass $u = \sum_{(i,j) \in \mathcal{U}} q_{ij}^0$ is positive. Give each such pair weight one except (h, x) , for one observed endpoint x , which receives weight two. Define G by retaining the observed rates, assigning unobserved rate $uw_{ij}/\sum_{\mathcal{U}} w_{ij}$, and normalizing rows. Then G is complete and has the same trace as Q_0 . Put $P_\epsilon = (1 - \epsilon)Q_0 + \epsilon G$. For all sufficiently small $\epsilon > 0$ it is complete and the full external exit rows of h, k differ. Indeed their difference is $(1 - \epsilon)v + \epsilon w$ with $w \neq 0$, so it is nonzero near zero whether v vanishes or not.

TABLE II. Local upper boundedness under unknown bidirected support and fixed state count. The observation column uses joint waiting laws for one resolved pair.

Source	Entrywise-generator neighborhood	Observation neighborhood
$N = 2$	Yes	Yes
$N = 3$, complete	Yes	Yes
$N = 3$, tree	No	No
$N \geq 4$, complete	Yes	No
$N \geq 4$, noncomplete	No	No

Let $T_0 = Q_0 - E$, where E is the sum of marked matrices. All marked matrices, R , and B stay fixed. Positivity of the killed semigroups, Duhamel's identity, and $\int_0^\infty e^{(P_\epsilon - E)t} B \mathbf{1} dt = \mathbf{1}$ give

$$d(\mathcal{O}(P_\epsilon), \mathcal{O}(Q_0)) \leq \frac{\epsilon}{2} \max_I [R(-T_0)^{-1} |G - Q_0| \mathbf{1}]_I. \quad (33)$$

The constant is finite. Appendix D gives an exact hidden-pair similarity orbit of each fixed P_ϵ that preserves its complete marked kernel and approaches a one-way limiting flux with positive opposing flux. Its entropy diverges. First choose ϵ to make (33) smaller than η , then choose a member of this fixed orbit with entropy above L . Similarity retains trace, and nonnegative rates sum to τ , giving the claimed common upper cap. The order of choices does not require any uniform lower bound on the divergence coefficient as $\epsilon \rightarrow 0$. $\square$

Proof of Theorem 5. Two-state chains have zero stationary current and entropy. At a complete three-state source, Theorem 1 gives local entropy regularity, hence a finite ceiling, in either data topology; restricting the class preserves it. Every three-state tree admits the missing-pair construction after Proposition 4. There is a positive unobserved edge to compensate, so observed rates and trace remain fixed. The same coupling used there gives row-TV convergence and divergent entropy. For $N \geq 4$, one observed pair leaves at least two fully hidden states, and Lemma 6 applies. TV convergence implies weak convergence, so these negative cases also exclude a weak neighborhood ceiling. $\square$

The cap supplied by the construction is τ , not every upper cap containing the source. For example, at $N = 4$, trace -12 and $q_{ij} \leq 1$ force all twelve rates to equal one. That restricted class is a singleton despite two fully hidden states. Known sparse topology and positive reverse-rate information can likewise change the conclusion.

VI. FINITE RECORDS AND STATISTICAL INFERENCE

Let $\mathbb{P}_Q^H$ denote the stationary observed-record law on $[0, H]$, including empty records and both censored ends. The divergent constructions above also give, at their negative-case sources,

$$\sigma(Q_n) \rightarrow \infty, \quad \text{TV}(\mathbb{P}_{Q_n}^H, \mathbb{P}_{Q_0}^H) \rightarrow 0 \quad \text{for each fixed } H > 0. \quad (34)$$

For the sparse sequence this follows from stationary-distribution continuity and common/excess-clock coupling. For the hidden-pair sequence, the completion $P_\epsilon \rightarrow Q_0$ has the same property, and each subsequent similarity preserves the stationary marked-record law exactly. To see the latter, use $\pi' = \pi S$, $S\mathbf{1} = \mathbf{1}$, and $SE_J = E_J S$ in each record density

$$\pi e^{Tt_1} E_{J_1} e^{T(t_2-t_1)} \dots E_{J_n} e^{T(H-t_n)} \mathbf{1}. \quad (35)$$

All similarities cancel, also for $n = 0$. For (27) a quantitative bound is

$$\text{TV}(\mathbb{P}_{Q(e,k)}^H, \mathbb{P}_{Q(0,0)}^H) \leq \frac{(e-k)(5+e-k)}{3Z} + (e+k)H \leq \frac{11e}{18} + 2eH. \quad (36)$$

These statements establish convergence in a distinct experiment, not by an assumed implication from row TV alone.

Corollary 7 (Uniform upper-confidence obstruction). *Suppose a model class contains a sequence (34). Let U be a measurable extended-real function of the stationary record on a fixed $H > 0$, possibly with independent randomization. If, for $0 < \alpha < 1$,*

$$\mathbb{P}_Q^H\{U \geq \sigma(Q)\} \geq 1 - \alpha \quad \text{for every model in the class,} \quad (37)$$

then $\mathbb{P}_{Q_0}^H\{U = +\infty\} \geq 1 - \alpha$.

Proof. For every finite L , eventually $\sigma(Q_n) > L$. Transfer the coverage inequality through TV convergence to obtain $\mathbb{P}_{Q_0}^H\{U > L\} \geq 1 - \alpha$. Let $L \uparrow \infty$ and use continuity from above. Adjoining an independent random seed leaves TV convergence unchanged. $\square$

This is an application of the established confidence-set impossibility mechanism [30], with the required physical sequences supplied here. It concerns uniform finite-sample coverage over the declared class. It does not settle pointwise asymptotics, Bayesian prior restrictions, or a horizon growing jointly with the sequence. We also do not infer a cap-free positive continuity theorem from one stationary finite window; Theorem 1 concerns complete joint waiting laws.

A. A quantitative rare-rate testing obstruction

The boundary family gives an exponential cost for a precisely specified testing problem. The asymptote in Eq. (29) implies

$$\frac{\log(1/k)}{\sigma(Q(e, k))^2} \longrightarrow 9, \quad k = \exp(-1/e^2). \quad (38)$$

This is a logarithmic equivalence, not the multiplicative claim $k \sim \exp(-9\sigma^2)$; subleading entropy terms matter inside an exponential.

Proposition 8 (Testing independently variable reverse rates). *Fix $p > 1$ and $0 < \gamma < 1$. Compare stationary records of $Q_1 = Q(e, k)$ and $Q_p = Q(e, k^p)$, where $k = \exp(-1/e^2)$ and $0 < e < 1/2$. If a test, even one observing the full microscopic trajectory on $[0, H]$, has sum of its two error probabilities at most γ , then*

$$H \geq \frac{1 - \gamma}{k - k^p} - 1. \quad (39)$$

The same necessary bound holds for the resolved-pair record. As $e \downarrow 0$, $\sigma(Q_p)/\sigma(Q_1) \rightarrow p$ and the logarithm of the right side divided by $\sigma(Q_1)^2$ tends to 9 (or $9/p^2$ when expressed using $\sigma(Q_p)^2$).

Proof. On $0 \leq k \leq e < 1/2$, the stationary weights in Proposition 4 give

$$\left\| \frac{\partial \pi}{\partial k} \right\|_{\text{TV}} \leq \frac{3 - 2k}{Z} \leq 1. \quad (40)$$

Indeed $Z = 3 + e(1 - e) + k(2 - e - k) \geq 3$, and applying the quotient rule to the three weights gives the first bound. Changing k to k^p changes one outgoing rate in each of rows y, h , with maximum common/excess-clock separation hazard $k - k^p$. Couple the stationary initial laws maximally and then the clocks. The full-path TV distance is at most $(1 + H)(k - k^p)$. Observation cannot increase this distance. The sum of testing errors is at least one minus TV, proving Eq. (39). Substitution into Eq. (28) yields $e\sigma(Q_p) \rightarrow p/3$. $\square$

Consequently, for every fixed $c < 9$ an eventual lower bound of the form $C \exp(c\sigma(Q_1)^2)$ follows, with $C > 0$ and a sufficiently small- e domain. Both alternatives retain the observed rates, trace and upper rate cap. This is a lower bound for separating alternatives with a fixed multiplicative entropy gap; it is not an achievable upper bound for estimating entropy from marked data. Rejection with prescribed error probabilities of a high-entropy alternative separated from the true entropy requires such a test.

There is also a direct rare-event diagnostic. If the reverse microscopic transition $h \rightarrow x$ were itself counted, stationarity gives

$$\mathbb{E}N_{h \rightarrow x}(H) = \pi_h k H, \quad \Pr\{N_{h \rightarrow x}(H) \geq 1\} \leq \pi_h k H. \quad (41)$$

Thus detection probability at least $1 - \beta$ requires $H \geq (1 - \beta)/(\pi_h k)$, with $\pi_h \rightarrow 1/3$. The observed-event intensity tends to $2/3$, so the corresponding expected observed-event count has the same exponential order. A stationary intensity is not an assertion that rare counts are exactly Poisson. The hidden reverse jump is unavailable to our actual detector; the full-path testing bound already accounts for that information loss.

The class assumption is decisive. If the exact relation $k = \exp(-1/e^2)$ is imposed as known prior information for *every* competitor, $Q(e, k^p)$ with $p > 1$ is excluded. Then estimating the more common entrance rate e can determine entropy through that prior, without directly detecting a reverse jump. Nor does comparing only $Q(e, k)$ with the limiting tree establish an exponential cost: the tree differs in rates of order e . The quantitative obstruction requires an independently variable small reverse rate, as allowed in the declared generator class.

B. Constructive finite-data inference

The finite-record obstruction does not preclude useful pointwise inference at a well-resolved interior model. Here we estimate all six rates from the observed pair alone. Discard the censored interval before the first mark and record n complete inter-mark intervals. This stops at a random horizon; it differs from the fixed-window experiment of Corollary 7. For incoming mark I_m and outgoing mark J_m , the empirical joint moments are

$$\hat{\mu}_{I,jL} = \frac{1}{n_I} \sum_{m: I_m=I} e^{-\lambda_j \tau_m} \mathbf{1}\{J_m = L\}, \quad n_I = \sum_m \mathbf{1}\{I_m = I\}. \quad (42)$$

We fit all twelve moments by nonlinear least squares over six positive log-rate coordinates, calculate the stationary entropy of the fitted generator, and estimate uncertainty by a sandwich delta method.

Successive mark rows are dependent. Nevertheless the centered row features

$$Z_m^{(I)} = \mathbf{1}\{I_m = I\}(X_m - \mu_I), \quad X_m = (e^{-\lambda_j \tau_m} \mathbf{1}\{J_m = L\})_{j,L}, \quad (43)$$

are bounded martingale differences: the incoming mark resets the known microscopic state. Cross-time covariances vanish, and the two row blocks have disjoint indicators at a common time. At

a fixed complete positive generator the bounded martingale central limit theorem and ergodicity give

$$\sqrt{n}(\hat{\mu} - \mu) \Rightarrow N(0, \Omega), \quad \Omega = \text{blockdiag}(C_I/p_I), \quad (44)$$

$$p_+ = \frac{\pi_x r}{\pi_x r + \pi_y s}, \quad p_- = 1 - p_+, \quad (45)$$

$$(C_I)_{jL, \ell M} = \mathbf{1}\{L = M\} \hat{\Psi}_{IL}(\lambda_j + \lambda_\ell) - \hat{\Psi}_{IL}(\lambda_j) \hat{\Psi}_{IM}(\lambda_\ell). \quad (46)$$

Replacing $C_I/(np_I)$ by the empirical within-row feature covariance divided by n_I gives a consistent estimate $\hat{C}$ of the moment covariance. This argument does not assume independent successive marks or independence conditional on the random row counts.

For a consistent interior fit, let $A = (J_{\log}^T J_{\log})^{-1} J_{\log}^T$ and $g_{\log} = \text{diag}(\theta) \nabla_{\theta} \sigma$. The estimated entropy variance is

$$\hat{V}_{\sigma} = g_{\log}^T A \hat{C} A^T g_{\log}. \quad (47)$$

An SVD implementation avoids forming the squared-condition normal equations. The interval $\hat{\sigma} \pm 1.96 \sqrt{\hat{V}_{\sigma}}$, intersected with $[0, \infty)$, has only pointwise asymptotic justification at an interior nonequilibrium truth. At equilibrium the first entropy derivative vanishes; near a sparse boundary the approximation need not be accurate. A global moment minimizer on a fixed compact positive rate box is consistent by exact injectivity and uniform criterion convergence. Our multistart numerical optimizer is not a certificate of global minimization.

For an explicit implementation we impose the numerical rate box $[10^{-14}, 20]$, use $\lambda = (0.2, 1, 5)$, three fixed starting points, analytic derivatives, and stopping tolerances 10^{-10} . The box is a stated computational restriction, not physical evidence for a reverse-rate floor. We flag estimates at its boundary or with unresolved logarithmic rank. A separate practical warning suppresses a suggested interval if either same-mark count is below 20; this threshold is a heuristic, not a coverage theorem. It addresses a concrete failure: zero observed ++ events give zero empirical variance for those moment features despite positive true variance when $b > 0$.

We simulated 100 independent stationary CTMC trajectories per model and sample size, for $n = 500, 5000, 20000$. Only resolved marks and their times enter the fit. The interior model is $(1.2, 0.8, 0.9, 0.4, 0.3, 1.1)$, with $\sigma = 0.20195$; the two sparse examples are $Q(0.2, 0.01)$ and $Q(0.2, e^{-25})$, with entropies 0.13143 and 1.17303. All true rates exceed the numerical lower bound. Table III reports all point estimates, including boundary fits, and the raw delta calibration before the additional same-mark warning. Missing intervals are not counted as successful coverage.

Model	n	Bias	Variance	RMSE	Interior	Covered
Interior	500	-0.00047	0.00271	0.0518	100	95
	5000	-0.00054	0.000317	0.0177	100	96
	20000	0.00098	0.0000869	0.00933	100	93
$b = 0.01$	500	0.0977	0.134	0.378	93	88
	5000	0.00152	0.000303	0.0174	100	94
	20000	-0.00007	0.0000620	0.00783	100	95
$b = e^{-25}$	500	-0.336	0.463	0.756	54	4
	5000	-0.315	0.371	0.683	55	9
	20000	-0.227	0.357	0.637	50	7

TABLE III. Monte Carlo inference from 100 trajectories in each cell. “Interior” counts successful nonboundary fits with resolved logarithmic rank; “Covered” counts their raw 95% delta intervals containing the truth. Dividing by the interior count gives conditional coverage; dividing by 100 also penalizes missing intervals. For example, the last row has 14% conditional coverage and 7% covered-and-interior fraction. A Wilson 95% interval for the Monte Carlo coverage is $[0.888, 0.978]$ for 95/100 and $[0.0695, 0.262]$ for 7/50. Bias and variance include every point estimate.

At the interior truth, 5000 records give 8.8% relative entropy RMSE; 20000 give 4.6% and a median interval width of 0.0343. The oracle asymptotic calculation predicts roughly 1.44×10^4 records for a 95% half-width of 10% of this entropy, which is a planning estimate rather than a finite-sample guarantee. At $b = 0.01$, 20000 records give 6.0% relative RMSE. All 900 selected optimizations converged, but this did not imply reliable entropy inference. For $b = e^{-25}$, 45–50% of fits reach the lower floor and lose numerical rank; the remaining raw intervals have only 7–16% conditional coverage. Every rare-case trajectory has zero microscopic $h \rightarrow x$ jumps and zero observed $++$ pairs. The former is a simulation diagnostic unavailable to inference; the latter suppresses all suggested intervals under the practical count warning. At 20000 records the median maximum fitted moment residual is only 5.8×10^{-4} , yet the entropy RMSE remains 0.64. Small observable residuals do not establish accurate rare-rate or entropy estimates.

This pipeline demonstrates useful inference in regular examples and exposes its failure when a reverse rate is unresolved. It does not establish uniform finite-sample coverage, an efficient estimator, or an optimal sample-complexity exponent. Weighted moments or a full marked-trajectory likelihood can change efficiency, while honest upper uncertainty near the singular boundary still requires explicit additional information.

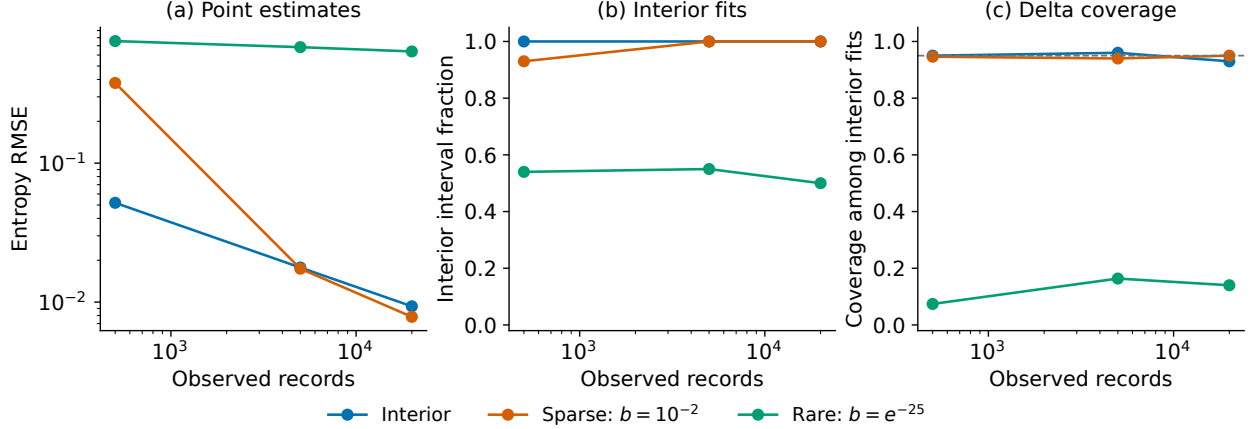


FIG. 3. Constructive inference and its failure near a sparse boundary. Each point uses 100 actual CTMC trajectories with the parameters in Table III. (a) Entropy RMSE includes boundary fits. (b) Fraction of fits permitting an interior delta calculation. (c) Raw interval coverage conditional on those fits, before the extra rare-mark warning; the dashed line is 95%. The rare-reverse case is poorly calibrated even after numerical boundary exclusion. The machine-readable output includes bias, variance, Wilson coverage intervals, all fitted rates, event counts, optimization statuses, and deterministic seeds.

C. True-law distance and finite-record confidence sets

The observation topology in the stability theorems is a distance between *true* row laws. It is not the total-variation estimation error of a raw empirical measure. For every finite nonempty row, $\hat{\mu}_I = n_I^{-1} \sum_{m:I_m=I} \delta_{(\tau_m, J_m)}$ is atomic, while the CTMC row law has a density in time. The finite set of observed atoms has empirical probability one and true probability zero, so

$$\text{TV}(\hat{\mu}_I, \mu_I) = 1. \quad (48)$$

Weak convergence and convergence of bounded Laplace tests remain possible. The moment estimator above uses precisely those bounded tests and therefore does not assume a shrinking empirical-TV error. A smoothing or detector model would need its own error analysis.

An elementary confidence rectangle gives a finite-sample alternative to the pointwise delta approximation.

Proposition 9 (Sequential moment confidence rectangle). *Fix K positive Laplace arguments in advance, $0 < \alpha < 1$, and a deterministic number $n \geq 1$ of complete inter-mark intervals. Let $d_{\text{mom}} = 4K$ and*

$$t_{n,\alpha} = \sqrt{\frac{n}{2} \log \frac{2d_{\text{mom}}}{\alpha}}. \quad (49)$$

For the empirical joint moments (42),

$$\mathbb{P}_Q \left\{ \left| \hat{\mu}_{I,jL} - \hat{\Psi}_{Q,IL}(\lambda_j) \right| \leq \frac{t_{n,\alpha}}{n_I} \text{ for all } (I, j, L) \text{ with } n_I > 0 \right\} \geq 1 - \alpha. \quad (50)$$

Rows with $n_I = 0$ impose no constraint. Stationarity of the initial mark is unnecessary; the exact CTMC and resolved-reset observation assumptions suffice.

Proof. Let $\mathcal{F}_0$ contain the first observed mark and let $\mathcal{F}_m$ include the next m complete marked intervals. Then I_m is $\mathcal{F}_{m-1}$ -measurable. For one fixed coordinate define

$$Z_m = \mathbf{1}\{I_m = I\} \left(e^{-\lambda_j \tau_m} \mathbf{1}\{J_m = L\} - \hat{\Psi}_{Q,IL}(\lambda_j) \right). \quad (51)$$

The reset property gives $\mathbb{E}[Z_m \mid \mathcal{F}_{m-1}] = 0$. Conditionally, Z_m lies in an interval of length at most one, so conditional Hoeffding's lemma yields $\mathbb{E}[e^{\eta Z_m} \mid \mathcal{F}_{m-1}] \leq e^{\eta^2/8}$. Iteration and Chernoff's bound, optimized at $\eta = 4t/n$, imply

$$\mathbb{P}_Q \left\{ \left| \sum_{m=1}^n Z_m \right| > t \right\} \leq 2e^{-2t^2/n}. \quad (52)$$

Union over d_{mom} coordinates and use $\sum_m Z_m = n_I(\hat{\mu}_{I,jL} - \hat{\Psi}_{Q,IL}(\lambda_j))$ for nonempty rows. The argument neither assumes independent successive rows nor conditions on their random counts. $\square$

Intersecting the simultaneous rectangle with a declared microscopic model class gives a confidence set for the generator with coverage at least $1 - \alpha$. Its full entropy image, or a certified enclosing interval, inherits this coverage and may have an infinite upper endpoint. Finding a large feasible entropy by numerical optimization does not certify that endpoint from above. The bound is conservative and does not establish efficient inversion or an efficient estimator. It applies to fixed n complete intervals and preselected arguments; an adaptive stopping rule, same-record design selection, terminal censoring at fixed physical time, or missed events requires additional analysis.

The existing simulations already use actual sequential CTMC records and dependence-aware moment covariance. Their reported bias, variance, and Monte Carlo coverage remain operationally meaningful. Their Wald intervals and low-count warning retain the stated pointwise or heuristic status; Eq. (50) is a separate finite-sample confidence-set construction, not a retroactive coverage guarantee for those intervals.

VII. PHYSICAL REGULARIZATION AND UPPER BOUNDS

Let

$$\mathcal{K}(Q) = \sum_i \pi_i \sum_{j \neq i} q_{ij}, \quad a_{\max}(Q) = \max_{q_{ij} > 0} \left| \log \frac{q_{ij}}{q_{ji}} \right|. \quad (53)$$

Here $\mathcal{K}$ is the microscopic activity, in inverse-time units; $a_{\max}$ is dimensionless. Stationary cancellation gives the elementary bound $\sigma \leq \mathcal{K}a_{\max}$. A sharper established steady-state bound is [29]

$$\sigma(Q) \leq \mathcal{K}(Q)a_{\max}(Q) \tanh\left(\frac{a_{\max}(Q)}{2}\right). \quad (54)$$

The right side is zero at $a_{\max} = 0$. Thus an activity ceiling $\mathcal{K} \leq K_0$ and an affinity ceiling $a_{\max} \leq a_0$ give a uniform finite entropy bound. An observed-event activity ceiling is not by itself a ceiling on microscopic activity. If all supported rates lie between $m > 0$ and M , then $\mathcal{K} \leq (N-1)M$ and $a_{\max} \leq \log(M/m)$; a fixed trace $-\text{tr } Q = \tau$ also gives $\mathcal{K} \leq \tau$.

A rate ceiling alone fails by Proposition 4. Conversely, an affinity ceiling alone does not control arbitrarily fast dissipative cycles. A rate floor alone also fails. In fact even a floor together with an activity ceiling is insufficient: take $q_{12} = c \geq 1$ and all five other rates of a complete three-state chain equal to one. Direct stationary solution gives

$$\pi = \left(\frac{1}{c+2}, \frac{2c+1}{3(c+2)}, \frac{1}{3}\right), \quad \mathcal{K} = \frac{3(c+1)}{c+2} < 3, \quad \sigma = \frac{c-1}{3(c+2)} \log c \longrightarrow \infty. \quad (55)$$

The fast rate acts at a rarely occupied state. These priors therefore control different quantities and cannot be interchanged.

There is a separate continuity statement at the generator level. On a fixed connected support with $m \leq q_{ij} \leq M$, $Q \mapsto \sigma(Q)$ is uniformly Lipschitz on the compact admissible class. With a fixed affinity bound, entropy is continuous under entrywise convergence to an irreducible generator even if reciprocal pairs disappear: contributions from pairs positive at the limit are smooth, and each disappearing directed term is bounded in magnitude by $a_0 \pi_i q_{ij} \rightarrow 0$. Locally the corresponding bounded-ratio cone even gives Lipschitz continuity, since the derivatives of $x \log(x/y)$ are bounded when x/y stays in a compact positive interval. These conclusions concern the entropy functional. They do not imply generator identification or good conditioning of a partially observed inverse, and a compact parameter class alone need not make entropy single valued on an exact fiber. If entropy is identifiable throughout such a compact class, its induced function on the observed image is continuous by the compact quotient argument; a quantitative inverse bound still needs additional analysis.

A. Compact physical priors and shrinking data tolerance

Fix a state cap $\bar{n} \geq 2$, the named microscopic endpoints of the resolved reverse pair, and $0 < m < M < \infty$. Let $\mathcal{C}_{\text{reg}}$ contain the connected reciprocal generators with $2 \leq N \leq \bar{n}$ for which

every supported rate belongs to $[m, M]$. Absent reciprocal pairs are allowed; the observed pair is present with its declared incidence. The values of its rates may be unknown. Equip this class with the finite disjoint-union topology over dimensions and the usual matrix topology within each dimension. Hidden-state relabeling does not change the conclusions.

Proposition 10 (Attained upper bounds under compact priors). *The class $\mathcal{C}_{\text{reg}}$ is compact, its joint waiting-law map is continuous in maximum row total variation, and its stationary entropy functional is continuous. For feasible exact data $D \in \mathcal{O}(\mathcal{C}_{\text{reg}})$ define*

$$U_D(\eta) = \max\{\sigma(Q) : Q \in \mathcal{C}_{\text{reg}}, d(\mathcal{O}(Q), D) \leq \eta\}, \quad \eta \geq 0. \quad (56)$$

Every maximum exists and is finite, and

$$U_D(\eta) \downarrow U_D(0) \quad (\eta \downarrow 0). \quad (57)$$

The assertion also holds with a metric generating weak convergence of the joint row laws in place of d .

Proof. For each fixed dimension and connected reciprocal support, supported ordered rates range over a closed finite box and diagonals are fixed by row sums. There are only finitely many such supports and dimensions. Their union is compact. In a converging fixed-dimensional sequence, each off-diagonal lies in $\{0\} \cup [m, M]$, so its support is eventually constant. The limit retains a connected support and is irreducible.

Here is an explicit common waiting-time tail. Set

$$\nu = (\bar{n} - 1)M, \quad t_0 = 1/\nu, \quad b = \frac{e^{-1}}{\bar{n}!} \left(\frac{m}{\nu}\right)^{\bar{n}} > 0. \quad (58)$$

Append an absorbing state reached at any observed jump and uniformize at rate ν . From every state, a path to absorption uses at most $\bar{n}$ transitions, each with uniformized probability at least m/ν . Absorption earlier on the path also counts. The event of exactly $\bar{n}$ Poisson steps in time t_0 therefore gives absorption probability at least b . Repeating the bound by the Markov property,

$$\sup_{Q \in \mathcal{C}_{\text{reg}}, i} \Pr_i^Q\{\tau_{\text{mark}} > t\} \leq (1 - b)^{\lfloor t/t_0 \rfloor}. \quad (59)$$

On a finite time interval, continuity of the matrix exponential gives uniform convergence of $Re^{T_Q t} B_Q$. Equation (59) controls the discarded probability tails uniformly. Splitting the row-TV integral at a large finite time proves continuity of $\mathcal{O}$. The normalized stationary solution and the entropy sum are continuous on each support box, because the stationary distribution and supported rates stay positive. Finite union gives continuity on $\mathcal{C}_{\text{reg}}$.

The constraint set in Eq. (56) is nonempty and compact, so its maximum is attained. These sets shrink with η , and their maxima are bounded below by $U_D(0)$. For any $\eta_j \downarrow 0$, extract a convergent subsequence of maximizers. Its limit belongs to the exact fiber by observation continuity; entropy continuity makes the limiting value at most $U_D(0)$. This proves Eq. (57). Row-TV continuity also implies weak continuity, so the same compactness argument applies to a metric for weak convergence. $\square$

An explicit data-independent bound follows by setting $A = \log(M/m)$. Microscopic activity is at most $(\bar{n} - 1)M$, so the established activity–affinity bound [29] gives

$$U_D(\eta) \leq (\bar{n} - 1)MA \tanh(A/2). \quad (60)$$

The prior bounds make the constrained upper value finite; they do not make the compatible generator or its entropy unique. The proposition holds the feasible center D fixed and shrinks its tolerance. It does not assert two-sided continuity as D varies, good inverse conditioning, or a global certificate for a numerical optimizer. Closed tolerance balls are used for attainment; an open ball may have only a supremum.

Finally, this deterministic statement is not a confidence-set construction. Raw empirical waiting laws are atomic and have row-TV distance one from the continuous model laws. A practical use therefore requires a justified fitted-density TV region or a weaker metric or measurement region with its own statistical coverage. The rate floor, rate ceiling and state cap remain explicit physical assumptions.

B. Physical interpretation and measurement limitations

The theorems concern the stationary network entropy-production functional of an even-state CTMC with one channel per ordered pair. They require no specified reservoir realization. Interpretation as physical dissipation requires correctly resolved thermodynamic states and channels obeying local detailed balance [31]. For a channel with intrinsic state entropies s_i , the rate ratio may take the form

$$\log \frac{q_{ij}^\alpha}{q_{ji}^\alpha} = \Delta s_{\text{res}}^\alpha + s_j - s_i. \quad (61)$$

At stationarity, total and medium entropy rates agree because the mean state-entropy increment vanishes. We use the usual environmental sign convention; the trajectory quantities differ by a boundary term. The generator alone does not determine a unique heat current or reservoir

decomposition. Aggregating parallel channels can conceal additional physical dissipation even when all state transitions are seen.

In the sparse-boundary construction, the vanishing reverse rate entails an unbounded edge rate affinity and hence an unbounded per-transition thermodynamic bias under local detailed balance. A fixed affinity prior excludes that limit. Rate ceilings and floors are kinetic assumptions; their ratio constrains the thermodynamic bias. A cycle affinity is the sum of edge rate affinities and is distinct from a bound on each edge.

Finally, timestamp bins, missed events, uncertain endpoints and blurred labels require an explicit observation model. Finite timing resolution with every event retained is different from detector dead time; the former is the setting of the finite-resolution comparison in Ref. [18]. A known noisy processing of the same records contracts total variation and cannot eliminate an existing indistinguishability obstruction. Such processing may also remove the reset structure needed for our reconstruction. For general coarse or semi-Markov observations, the time reverse must be induced from the microscopic experiment; a naive reversal can create spurious irreversibility when coarse-graining and time reversal do not commute [24, 25].

VIII. AN EXACTLY TWO-POINT ENTROPY FIBER

The following family has unbounded subcritical entropy and exactly classified critical and supercritical fibers within a single fixed-dimensional inverse problem. Rates are expressed in the time unit of Sec. II. Linear minimality means that the columns of $T^j B$ span the state space and that $RT^j v = 0$ for every $j \geq 0$ forces $v = 0$. The minimal linear order is the least dimension of any matrix realization of the kernel; physical positivity is an additional constraint.

Theorem 11 (Balanced five-state fiber). *In state order (x, y, h, k, l) , let*

$$Q(z) = \begin{pmatrix} -11 & 1 & 2 & 0 & 8 \\ 2 & -20 & 0 & 7 & 11 \\ 3 & 0 & -7 & 4 & 0 \\ 0 & 6 & 5 & -11 & 0 \\ z & z & 0 & 0 & -2z \end{pmatrix}, \quad z > 0. \quad (62)$$

Let z_ be the unique positive root of*

$$\mathcal{P}(z) = 352z^4 - 3168z^3 - 4644z^2 - 11340z - 7623. \quad (63)$$

Every $\Psi_{Q(z)}$ has minimal linear realization dimension five. In $\mathcal{C}_{\leq 5}$ its entropy fiber is unbounded for $0 < z < z_*$, contains exactly two distinct finite values at $z = z_*$, and is a singleton with a unique physical generator for $z > z_*$. At z_* , the two generators are $Q(z_*)$ and the algebraic generator Q_* defined in Appendix C. Their entropy values satisfy the rational enclosures

$$0.062946400236 \leq \sigma(Q(z_*)) \leq 0.062946400237, \quad (64)$$

$$0.065579685966 \leq \sigma(Q_*) \leq 0.065579685967. \quad (65)$$

Here $z_* = 10.5571496686650 \dots$ is an illustrative decimal approximation.

The assertion at z_* is stronger than exhibiting two observationally equivalent models. It excludes every third physical generator under the cap, including any on a different bidirected support. The source has six reciprocal edges; the alternative has seven and a connected hidden path (Fig. 4). Thus hidden relabeling cannot explain their difference. Above the threshold even the physical generator is unique, although its observation will have no finite local entropy upper ceiling by Theorem 5.

Proof structure. The marked kernel fixes the visible block and the hidden transfer $X(\lambda I - H)^{-1}Y$

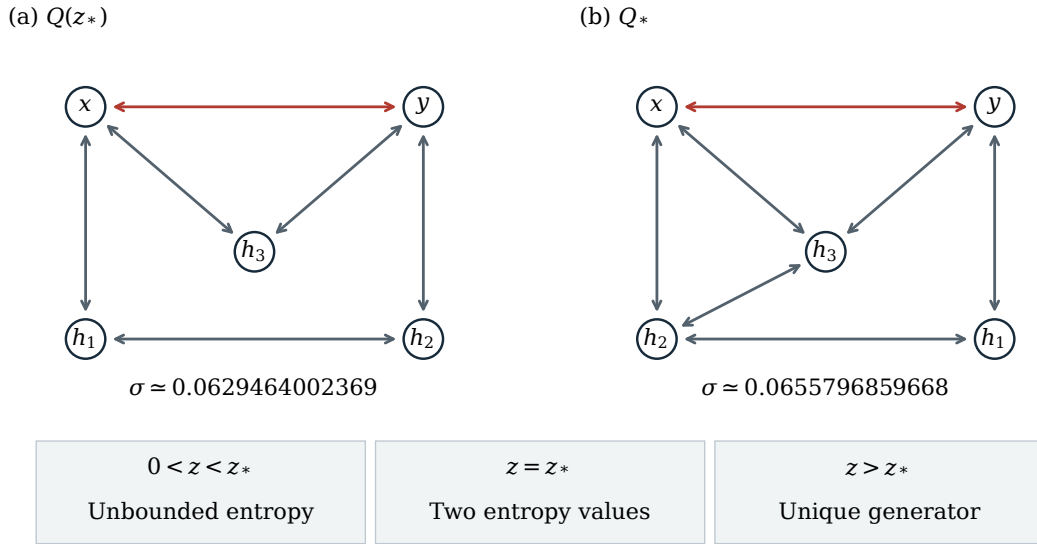


FIG. 4. Entropy-set properties for the family (62) under a five-state cap. Below the threshold the strip asserts unboundedness, not a complete range characterization. At the algebraic threshold the source and the connected-hidden alternative are the only physical realizations up to hidden labels. Paired arrows denote reciprocal support, not equal rates; only $x \leftrightarrow y$ is observed. Entropy values in the figure are rounded; Eqs. (64)–(65) give certified intervals.

Hidden components	Labeled supports	Exhaustive treatment for $z \geq 21/2$
Three singletons	1	A singleton cannot supply the mixed-sign residue at $-\beta$.
Pair and singleton	3	Singleton $-\beta$ is excluded by its residue; singleton $-\alpha$ by the pair's Perron residue. Fast singleton $-2z$ forces exactly the source rates.
Connected triple	4	Three paths and one triangle all enter the same invariant-triangle inequalities. No connected realization exists above z_* ; exactly one exists at z_* , after normalization and the separate feasibility certificate.

TABLE IV. Every reciprocal support on the three hidden vertices. Visible attachments are unrestricted initially and are constrained within each proof branch. Labels are counted before quotienting by hidden permutations. Below $21/2$ the theorem uses constructive unbounded paths, not this exclusion argument.

for the partition $Q = \begin{pmatrix} A & X \\ Y & H \end{pmatrix}$. Minimality and the state cap force every competitor into a normalized hidden-similarity orbit; Appendix A proves this reduction. A disconnected hidden support can realize the fast-parameter data only as the source, by a residue-sign and reciprocity argument (Appendix B). Every connected hidden support gives a finite invariant triangle after Perron normalization. Global containment and inward-flow inequalities reduce feasibility to a quadratic whose discriminant is a negative positive-factor multiple of $\mathcal{P}(z)$. Above z_* this excludes every connected realization. At z_* , equality forces a single triangle and hence at most one normalized generator. Appendix C independently constructs that generator, verifies physical feasibility, and separates its entropy from the source using rigorous logarithm bounds. Finally, Appendix E covers all $0 < z < z_*$ by exact positive realizations with a hidden pair whose entropy-divergent similarity orbit is proved in Appendix D. These arguments exhaust the generator classes at and above the threshold and prove unboundedness below it. They do not determine the full subcritical entropy set. $\square$

There is a modest strengthening of the subcritical assertion. For each fixed $0 < z < z_*$, the continuous exact-law path in Appendix D starts at a finite entropy value s_z and tends to $+\infty$. The intermediate value theorem gives

$$[s_z, \infty) \subseteq \mathcal{S}_{\mathcal{C}_{\leq 5}}(\mathcal{O}(Q(z))). \quad (66)$$

This is an attainable half-line for some finite s_z , not a claim about the full set, its infimum, or possible gaps below s_z .

The cap has physical content. It excludes additional nonminimal states invisible to the linear realization. Splitting one fully hidden state of (62) into two strongly lumpable copies preserves the complete marked kernel and permits entropy divergence on their internal edge (Appendix F). Thus one additional state makes the endpoint entropy fiber unbounded. Conversely, the cap can be replaced by requiring all competing realizations to be linearly minimal. No separate upper rate cap is needed in the exact fiber: similarity fixes $-\text{tr } Q = 49 + 2z$, which bounds each nonnegative off-diagonal rate. We make no general classification claim for arbitrary five- or six-state data.

A. Persistence under a transverse perturbation

The algebraic boundary of the balanced example is not destroyed by a small asymmetry of the fast hidden state's exits. Define $Q(z, \delta)$ from Eq. (62) by replacing its last row with

$$(z + \delta, z - \delta, 0, 0, -2z), \quad z > |\delta|. \quad (67)$$

The hidden pole locations and output wedge remain fixed as δ varies; the fast-pole input residue changes. This perturbation is neither a time rescaling nor a hidden permutation.

Proposition 12 (Local persistence of the feasibility transition). *In the reciprocal irreducible class with at most five states, there is a smooth critical curve $z_*(\delta)$ through $(z_*, 0)$. In a neighborhood of that point the compatible entropy fiber is unbounded for $z < z_*(\delta)$, has exactly two distinct finite values at $z = z_*(\delta)$, and has a unique generator for $z > z_*(\delta)$, up to hidden permutations. The curve satisfies*

$$\begin{aligned} \mathcal{P}_\delta(z) &= \mathcal{P}(z) - (20160z + 14112)\delta \\ &\quad - (352z^2 - 3168z + 15768)\delta^2 = 0, \\ z'_*(0) &= \frac{20160z_* + 14112}{\mathcal{P}'(z_*)} \simeq 0.4650042. \end{aligned} \quad (68) \quad (69)$$

This is a local statement for the specified perturbation, not a generic classification of five-state networks.

Proof structure. Appendix G proves the perturbed discriminant identity, certifies the strict support-exhaustion and construction margins on a rational rectangle, and transfers the uniqueness and subcritical opening arguments. The entropy gap persists locally by the balanced exact certificate and continuity on fixed positive supports. $\square$

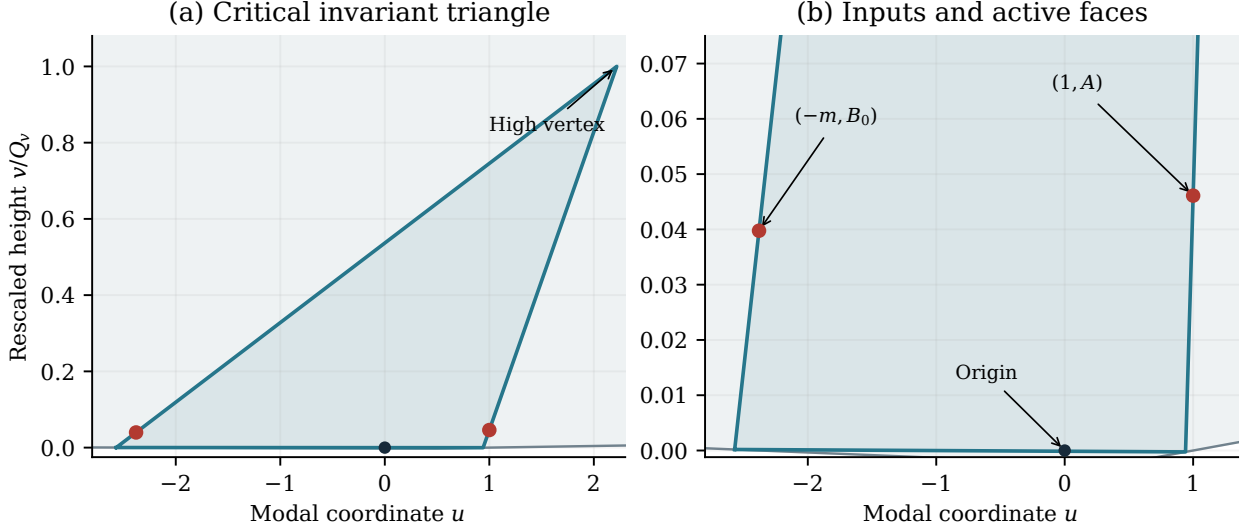


FIG. 5. Geometry at the critical balanced parameter. The invariant triangle lies inside the output wedge and contains the two input points (red). The vertical coordinate has been rescaled by its maximum vertex height; this invertible rescaling preserves incidences and invariance. Panel (b) enlarges the input region. Active input, output and tangent constraints determine the reciprocal path realization.

The reduced boundary equation has the exact normal form

$$F(t; z, \delta) = f_2(t - t_c)^2 - \frac{\Delta}{4f_2}, \quad t_c = -\frac{f_1}{2f_2}, \quad f_2 < 0. \quad (70)$$

Its two active-boundary path roots merge with square-root scaling as $\Delta \downarrow 0$. This is a nondegenerate fold of the boundary equation. The full feasible set has a complete-realization continuum below the curve and the source persists on both sides; it is therefore not a saddle-node of two isolated positive realizations. No conclusion about arbitrary perturbations of the source rates or hidden dimension follows from this one unfolding.

The deterministic continuation uses 41 δ nodes at spacing 0.025, 80-digit arithmetic, and offsets $z - z_*(\delta) = \pm 0.001$. At all these nodes it finds two, one, and zero admissible *active-boundary path candidates* below, at, and above the curve. These are not counts of all positive realizations outside the proved rectangle. Complete positive realizations are separately constructed below each node. The generator orbit Jacobian has numerical rank six, whereas the six active-zero constraint Jacobian has rank six on either path branch and rank five at their merger, using relative singular-value tolerance 10^{-10} . Present critical rates remain above 0.1645 in this grid. Structural zeros are checked below 10^{-65} without clipping; exact all-time equality comes from the modal similarity identity. Independent 120-digit recomputation at $\delta = -0.5, 0, 0.5$, direct matrix-exponential checks, and the historical exact path-witness comparison are recorded with source hashes, commands,

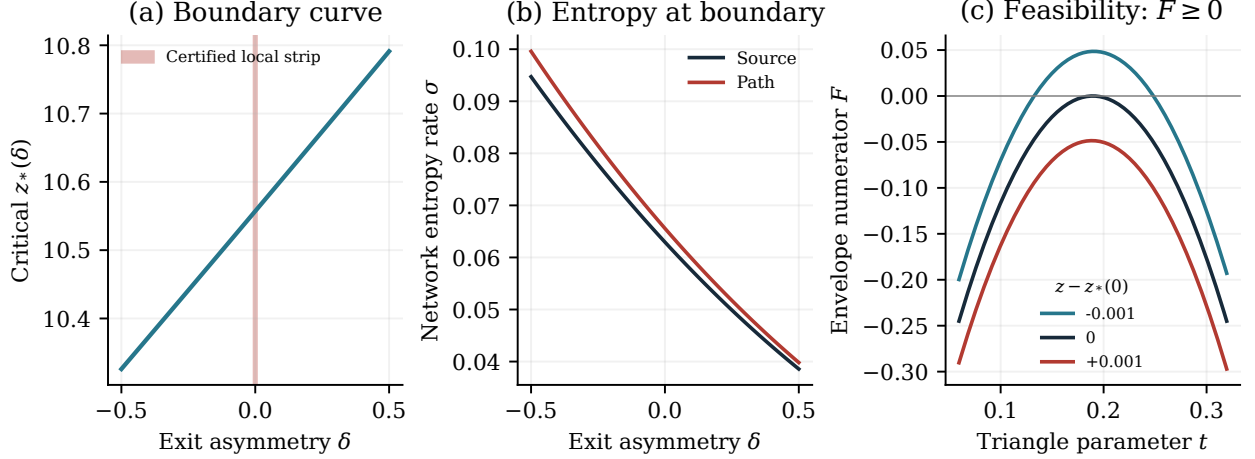


FIG. 6. Specified exit-asymmetry unfolding. (a) Numerical continuation of the boundary over $-0.5 \leq \delta \leq 0.5$; the narrow highlighted strip is the range with the exact geometric certificate (G5). (b) Source and path entropy at the numerical boundary. (c) The envelope numerator is nonnegative precisely between its two boundary roots below the critical parameter. The wider curves are numerical diagnostics, not global off-balanced support exhaustion.

precision and all numerical outputs.

IX. OBSERVATION GRAPHS BEYOND ONE RESOLVED PAIR

A. Coverage of microscopic vertices

Let $\mathcal{O}$ be a nonempty reverse-closed set of individually resolved microscopic transitions on a fixed, known vertex set V . Each observed transition has a positive rate and known endpoints. The observation graph has vertices V and the undirected pairs underlying $\mathcal{O}$ as edges. A vertex is covered if it is incident to an observed edge; an uncovered vertex is fully hidden. Writing E for the sum of observed jump matrices, we retain $T = Q - E$, $R_I = e_{\text{dest}(I)}^T$, and $B_J = q_J e_{\text{orig}(J)}$. The joint kernel is $Re^{Tt}B$.

Proposition 13 (Recovery under full vertex coverage). *If every microscopic vertex is covered, two distinct positive Laplace arguments determine the generator uniquely on the declared vertex set, even when the observed rates are unknown. The inverse on admissible data is rational and locally Lipschitz. It is therefore locally Lipschitz in maximum row total variation and continuous under weak row-law convergence, without a common rate cap. One positive argument suffices if all observed rates are known.*

Proof. For each vertex i , choose an observed mark I_i ending at i and an observed mark J_i beginning at i . Put $D = \text{diag}(q_{J_i})$ and select the square submatrix $F(\lambda)_{ij} = \widehat{\Psi}_{I_i, J_j}(\lambda)$. Then

$$F(\lambda) = (\lambda I - T)^{-1} D. \quad (71)$$

For $M_k = F(\lambda_k)^{-1}$ and $\lambda_1 \neq \lambda_2$,

$$M_2 - M_1 = (\lambda_2 - \lambda_1) D^{-1}, \quad D = (\lambda_2 - \lambda_1) (M_2 - M_1)^{-1}, \quad T = \lambda_1 I - D M_1. \quad (72)$$

For an additional observed mark J , let f_J be its measured column restricted to rows I_i . The identity $(\lambda_1 I - T) f_J = q_J e_{\text{orig}(J)}$ recovers its rate. Thus $Q = T + E$ is determined. The inverses have nonzero denominators at every admissible source. They remain smooth in a neighborhood, and the Laplace measurements are bounded continuous tests of the row laws. When the observed rates are known, D and E are known and the last formula in (72) needs only one argument. $\square$

At a complete source, Proposition 13 also gives a locally Lipschitz entropy inverse and a finite local upper ceiling. The same is true at an irreducible sparse source if competitors have the same known support. This condition requires fewer observations than all physical edges: observing $1 \leftrightarrow 2$ and $2 \leftrightarrow 3$ covers a complete three-state network while leaving $1 \leftrightarrow 3$ hidden. One unobserved reciprocal pair is therefore not sufficient for unbounded entropy inference.

The support condition remains consequential. Start from the unit tree $1 \leftrightarrow 2 \leftrightarrow 3$ and observe both tree edges. Open the unobserved pair at rates $q_{13} = e$ and $q_{31} = \exp(-1/e^2)$, keeping all four old rates fixed. The generators and their joint row laws converge to the tree, whereas the new pair contributes $\sigma \sim 1/(3e)$. Full vertex coverage identifies every generator in the sequence but does not make entropy continuous across this newly opening support. This variant changes the trace by a vanishing amount; a trace-preserving version needs unobserved rate slack.

At the other extreme, the existing hidden-pair construction applies whenever two vertices are uncovered: its similarity fixes every observed jump matrix, including when many edges are observed. Unknown support then allows unbounded entropy in every joint-kernel neighborhood at fixed dimension, observed rates and trace. These are two sufficient graph criteria; they do not exhaust arbitrary observation graphs with exactly one uncovered vertex. Nor do they extend directly to lumped states or blurred transition labels. A macrostate event need not reset to a known microscopic state, and consecutive waiting laws then need not determine the entire observed path law [16, 22, 26].

X. DISCUSSION AND LIMITATIONS

The local-ceiling theorem separates three sources of difficulty. Exact observation can admit distinct physical generators; absolute-rate inversion can be poorly conditioned at an unfortunate observation scale; and even a regular rate inverse need not make entropy continuous near a sparse boundary. The numerical inference pipeline addresses pointwise estimation within a declared positive rate box. It supplies neither a uniformly valid finite upper confidence limit over the unrestricted class nor a universal sample-size prescription.

Dimension is essential. The five-state classification applies to a known cap or a justified minimality restriction. A sixth state makes its exact fiber unbounded. When the dimension is unknown, the observed linear order supplies a lower bound on realization size, not an upper bound on physical complexity. Additional microscopic information is needed to exclude nonminimal hidden channels. The perturbed feasibility calculation supports a local continuation statement for a specified family, not typicality under arbitrary rate or observation perturbations. Arbitrary five- and six-state exact fibers remain unclassified.

The observation model assumes a stationary, time-homogeneous, finite-state CTMC, exact event times, known microscopic endpoints and correctly resolved transition labels. The waiting-law kernel describes the law following an event; a fixed stationary window includes both censored ends and has its own likelihood. Finite bandwidth, timestamp rounding, detector noise and missed events change the experiment. Missed events change the waiting kernel; detector memory, blurred labels or uncertain event identity can also invalidate the reset model. An accurately resolved microscopic jump still resets its endpoint even if earlier jumps were missed. More samples do not repair an incorrect detector model. Uncertain labels or lumped states require a different observation map. Finite temporal resolution and lost events must not be conflated.

Nonstationarity, continuous hidden degrees of freedom, memory and non-Markovian kinetics can invalidate the finite-dimensional representation. Local detailed balance, reservoir temperatures or chemical work, state parity under time reversal, and thermodynamic channel resolution are additional physical assumptions. We therefore distinguish the stationary network entropy functional from a claim of fully measured physical dissipation. Upper rate information is kinetic; controlling affinities constrains reservoir entropy only under those physical assumptions.

The exact conventional proofs, computational certificates and literature comparison support a focused theoretical contribution. They do not certify literature-wide priority or substitute for expert review of the global support exhaustion. A matching physical-realization theorem, a concrete

proof objection, or experimentally justified support, affinity or dimension information could change the appropriate claim or inference class.

XI. COMPUTATIONAL DETAILS AND REPRODUCIBILITY

The universal claims are established by the proofs in this article. Computation verifies exact identities, admissibility, and the rational entropy enclosures; numerical plots illustrate proved limits. There is no external dataset and no numerical optimization or graph enumeration used as a nonexistence proof.

OpenAI Codex, a GPT-6-based research and coding assistant, was used for literature retrieval, derivation and code preparation, manuscript drafting, and separate mathematical, citation, and reproducibility reviews. The checks below identify the executed validation; agreement between assistant reviews is not treated as independent scientific evidence. Expert human review remains pending.

The supplied package contains the source code and accepted outputs required for these checks. Exact arithmetic uses rational numbers and the real field $\mathbb{Q}(z_*)$, with $\mathbb{Q}(\sqrt{6})$ for the cone. Signs are evaluated in a specified real embedding through rational isolating intervals, not from leading polynomial coefficients. The endpoint checker validates all denominator exclusions, support zeros, positive rates, normalization, stationarity, and ten exact Markov-parameter matrices. Its independent cone and physical-path constructions agree coefficientwise. Appendix C specifies the logarithm remainder bounds.

The deterministic reproduction driver replays the principal symbolic certificates with output writes redirected, preserving historical artifacts. It independently checks the three-state stationary weights, cycle current, trace, inverse identities, and deterministic high-precision values. The original certificate figures use fixed parameter grids. New conditioning, continuation and Monte Carlo figures are numerical results with explicit parameter choices, precision, deterministic seeds, optimizer settings and tolerances. They are stored separately from exact proof-support outputs; finite grids do not establish genericity or uniform coverage. The accompanying manifest records commands, dependency versions, source hashes, and exact checks versus numerical diagnostics. Discovery searches preserved in the research repository are historical leads and are unnecessary for the certificates. The compact-prior and finite-sample moment propositions are analytical compactness and martingale proofs; symbolic and simulation outputs are not certificates of those arguments. New exact checks independently verify the global-envelope branch identities, two-point reconstruc-

tion and full-matrix counterexamples. No result is claimed formally verified; the repository’s Lean scaffold is not a verification of these theorems.

The root-level command `python reproduce_all.py` regenerates the essential exact checks, inference diagnostics, continuation and publication figures. The optional build flag compiles this manuscript. The finite-data and continuation implementations include tests against independent derivatives, exact transforms or algebraic identities. Source hashes and machine-readable outcomes distinguish completed runs from illustrative or exploratory evidence. All historic accepted scientific payloads are retained.

XII. CONCLUSION

Exact recovery of partially observed Markov dynamics does not guarantee a stable or finite upper inference of microscopic entropy production. With one resolved reverse pair and unknown reciprocal support, the fixed-count local-ceiling criterion is sharp. Statistical conditioning, rare reverse rates, model dimension and physically justified activity–affinity restrictions determine what that exact theorem permits an experiment to infer. The five-state two-point fiber is an explicit exhausted construction within this broader distinction, not a generic template for every hidden network.

ACKNOWLEDGMENTS

This draft was prepared with substantive assistance from OpenAI Codex for research auditing, mathematical and computational checking, literature verification, and manuscript composition. This disclosure does not assign authorship or replace expert human verification and responsibility. Author names, affiliations, funding, and the final disclosure require completion by the human authors before submission.

Data availability. All inputs are specified analytic generators. The accompanying reproducibility package supplies the LaTeX sources, figure scripts, exact-check scripts, and machine-readable outputs. The authors should select the final public archive or other journal-compliant access arrangement before submission; no public deposit is asserted in this draft.

Appendix A: Observation equivalence and minimum order

A matrix realization (R, T, B) is minimal if its reachable space $\text{span}\{T^j Bv : j \geq 0\}$ is the whole state space and its unobservable space $\{v : RT^j v = 0 \text{ for all } j \geq 0\}$ is zero. These are linear conditions; they impose no sign restriction on a change of realization coordinates.

Lemma 14 (Finite equality certificate and pinned similarity). *Realizations of dimensions n, n' have the same all-time kernel if and only if $RT^j B = R'T'^j B'$ for $0 \leq j < n + n'$. Equivalent minimal realizations have equal dimensions and a unique similarity intertwining their input, generator and output maps. For the resolved reverse pair in Sec. II, after fixing the visible order this similarity has the form*

$$S = \text{diag}(I_2, U), \quad U\mathbf{1} = \mathbf{1}, \quad Q' = S^{-1}QS. \quad (\text{A1})$$

Conversely, every such similarity yielding an admissible generator preserves the complete kernel. The entries of U need not be nonnegative.

Proof. Apply Cayley–Hamilton to $\text{diag}(T, T')$. Vanishing of the first $n + n'$ differences forces every subsequent difference to vanish, and the convergent exponential series gives all-time equality. Differentiation at zero proves the converse.

For minimal realizations define

$$S \left(\sum_j T'^j B' v_j \right) = \sum_j T^j B v_j. \quad (\text{A2})$$

If the vector on the left is zero, equality of Markov parameters makes every RT^ℓ applied to the vector on the right zero; source observability makes that vector zero. The same argument in reverse gives an inverse. Reachability then gives $SB' = B$, $ST' = TS$, $RS = R'$, and uniqueness. A sufficiently large block Hankel matrix $(RT^{i+j}B)_{i,j}$ has rank equal to the minimal dimension. Every realization of the same kernel factors this matrix through its state space, so a competing dimension cannot be smaller.

The zero-time data determine $q_{xy} = \psi_{-,+}(0)$ and $q_{yx} = \psi_{+,-}(0)$. After putting the same two visible states first, the reset maps coincide and the columns of $B = B'$ are positive multiples of the two visible coordinate vectors. Thus $SB = B$ pins the visible columns of S , and $RS = R$ pins its visible rows. This proves the block form of S .

The killed generator T is invertible. Indeed, in a finite irreducible chain each initial state has a positive probability of producing an observed jump in some common finite time. Taking the

minimum over the finitely many states gives a strictly positive probability. The Markov property then bounds survival geometrically over successive intervals, so $e^{Tt} \rightarrow 0$ and T is Hurwitz. Since $T\mathbf{1} = -B\mathbf{1}$,

$$TS\mathbf{1} = ST'\mathbf{1} = -SB\mathbf{1} = -B\mathbf{1} = T\mathbf{1}, \quad (\text{A3})$$

and invertibility gives $S\mathbf{1} = \mathbf{1}$. The observed jump matrices have nonzero entries only in the fixed visible block, so they commute with S . Consequently the similarity for T is also the stated similarity for Q . Direct substitution in $Re^{Tt}B$ proves sufficiency. $\square$

For the balanced source the hidden blocks are

$$X = \begin{pmatrix} 2 & 0 & 8 \\ 0 & 7 & 11 \end{pmatrix}, \quad Y = \begin{pmatrix} 3 & 0 \\ 0 & 6 \\ z & z \end{pmatrix}, \quad H = \begin{pmatrix} -7 & 4 & 0 \\ 5 & -11 & 0 \\ 0 & 0 & -2z \end{pmatrix}. \quad (\text{A4})$$

Writing Y_j for its columns, direct determinants give

$$\det(Y_1, Y_2, HY_1) = -9z(4z - 9), \quad (\text{A5})$$

$$\det(Y_1, Y_2, HY_2) = -18z(2z - 3), \quad (\text{A6})$$

$$\det \begin{pmatrix} X_{1,:} \\ X_{2,:} \\ (XH)_{2,:} \end{pmatrix} = -14(22z + 19). \quad (\text{A7})$$

For $z > 0$ the first two cannot vanish simultaneously and the last never vanishes. Hence (H, Y) is reachable and (X, H) observable, both with hidden dimension three.

To pass from hidden to full minimality, write $T = \begin{pmatrix} A_{\text{kill}} & X \\ Y & H \end{pmatrix}$. The columns of B span the visible plane. Applying T to that plane and subtracting the visible component puts $(0, Y_j)^T$ in the full reachable space. If $(0, h)^T$ is in that space, applying T and subtracting its visible component puts $(0, Hh)^T$ there as well. Hidden reachability thus gives full reachability. If a full vector is unobservable, $Rv = 0$ first forces its visible coordinates to vanish. Successive output equations then give $XH^j v_H = 0$ for every j , forcing $v_H = 0$ by hidden observability. Therefore the full minimal order is five for every $z > 0$, including repeated hidden poles and $z = 2$. The cap makes every competitor exactly five dimensional and minimal.

The observed rates and first derivatives determine the visible generator block A , including the killed visible diagonals. After the known row permutation and rate division, the observed Laplace

matrix determines the visible principal block $K(\lambda)$ of $(\lambda I - T)^{-1}$. For $\lambda > 0$, both $\lambda I - H$ and $\lambda I - T$ are invertible, and the Schur complement gives

$$X(\lambda I - H)^{-1}Y = \lambda I - A_{\text{kill}} - K(\lambda)^{-1}. \quad (\text{A8})$$

Thus every competitor has the same matrix-valued hidden transfer. Fitting its scalar entries separately would discard this common realization constraint.

Appendix B: Global support exhaustion and the invariant triangle

1. Modal coordinates and every disconnected hidden support

Put $\rho = \sqrt{6}$, $\alpha = 9 - 2\rho$, $\beta = 9 + 2\rho$, $r_y = 4(\rho - 1)/5$, and

$$D = \text{diag}(-\alpha, -\beta, -2z), \quad X_m = (p, q, v), \quad Y_m = \begin{pmatrix} r \\ f \\ w \end{pmatrix}, \quad (\text{B1})$$

$$p = \begin{pmatrix} 3 + \rho/2 \\ 35\rho/8 \end{pmatrix}, \quad q = \begin{pmatrix} 3 - \rho/2 \\ -35\rho/8 \end{pmatrix}, \quad v = \begin{pmatrix} 8 \\ 11 \end{pmatrix}, \quad (\text{B2})$$

$$r = (1, r_y), \quad f = (1, -4(\rho + 1)/5), \quad w = (z, z). \quad (\text{B3})$$

Direct multiplication gives

$$X(\lambda I - H)^{-1}Y = \frac{pr}{\lambda + \alpha} + \frac{qf}{\lambda + \beta} + \frac{vw}{\lambda + 2z}. \quad (\text{B4})$$

Here pr and vw are strictly positive rank-one residues, whereas qf has negative off-diagonal entries. In this appendix $z \geq 21/2$, so all three poles are distinct and $2z > \beta > \alpha > 0$.

Reciprocity makes hidden connected components the irreducible blocks of H' . On three hidden states there are precisely three component-size possibilities: $(1, 1, 1)$, $(2, 1)$, and (3) . In a disconnected case, each eigenvalue belongs to exactly one component, since the spectrum is simple. Every mode has its displayed nonzero transfer residue, by minimality. A singleton contributes a nonnegative outer-product residue. Three singletons, or a singleton at $-\beta$, are consequently impossible. If the singleton were at $-\alpha$, the remaining irreducible pair would have dominant eigenvalue $-\beta$. Its transfer residue there is $(X'r_P)(\ell_P^T Y')/(\ell_P^T r_P) \geq 0$ by its positive Perron vectors, again contradicting qf . Thus the singleton must have eigenvalue $-2z$.

The remaining pair has zeroth transfer moment $pr + qf = \text{diag}(6, 42)$. For either of its states, reciprocal visible support makes a positive off-diagonal contribution equivalent to attachment to

both visible states. Nonnegativity therefore forbids a shared visible neighbor. The two positive diagonal moments require one state attached only to x and the other only to y ; neither can lack visible attachment. Denote their entrance rates by x_1, x_2 , exits by y_1, y_2 , and internal rates by h_0, k_0 . The zeroth and first moments give

$$x_1 y_1 = 6, \quad x_2 y_2 = 42, \quad y_1 + h_0 = 7, \quad y_2 + k_0 = 11, \quad (\text{B5})$$

$$x_1 h_0 y_2 = 48, \quad x_2 k_0 y_1 = 105. \quad (\text{B6})$$

All denominators in the following elimination are positive physical rates. Multiplying the off-diagonal equations and dividing by the two diagonal equations gives $h_0 k_0 = 20$. Dividing the first off-diagonal equation by $x_1 y_1 = 6$ gives $h_0(11 - k_0) = 8(7 - h_0)$, or $19h_0 - h_0 k_0 = 56$. Hence

$$(h_0, k_0, y_1, y_2, x_1, x_2) = (4, 5, 3, 6, 2, 7). \quad (\text{B7})$$

The fast singleton's residue fixes its entrance/exit product; its exit sum $2z$ removes the remaining scalar freedom and gives entrances $(8, 11)^T$ and exits (z, z) . This exhausts all disconnected hidden supports and leaves exactly the source, up to hidden permutations.

2. Connected supports produce finite nondegenerate triangles

For a connected hidden support, the hidden block similarity provides a real invertible V such that

$$H' = V^{-1}DV \text{ is irreducible Metzler,} \quad X' = X_m V \geq 0, \quad Y' = V^{-1}Y_m \geq 0. \quad (\text{B8})$$

The Perron right eigenvector $b = V^{-1}e_1$ has a common strict sign. Since $X'b = p > 0$, that sign is positive. The left Perron eigenvector $a_P^T = e_1^T V$ also has a common strict sign; $a_P^T b = 1$ makes it positive. Write $s_i = (a_P)_i > 0$ for the first coordinate of column i of V . Sectioning the invariant cone $V\mathbb{R}_+^3$ at first coordinate one therefore gives exactly the convex hull of the three columns $V_{:,i}/s_i = (1, u_i, v_i)^T$. Their affine independence follows from $\det V \neq 0$. Moreover

$$e_1 = \sum_i b_i s_i \frac{V_{:,i}}{s_i}, \quad b_i s_i > 0, \quad \sum_i b_i s_i = 1, \quad (\text{B9})$$

so the planar origin lies strictly inside this finite triangle. This argument covers arbitrarily ill-conditioned but invertible similarities; it assumes no common compactness bound. A zero-first-coordinate ray is impossible in this connected branch, and disconnected supports have already been handled separately.

Since $e^{Dt}V = Ve^{H't}$ and $e^{H't} \geq 0$, the cone is invariant. Sectioning after division by the positive first-coordinate contraction, and rescaling time by $\beta - \alpha$, gives the planar flow

$$(u, v) \mapsto (e^{-t}u, e^{-kt}v), \quad k = \frac{2z - 9 + 2\rho}{4\rho} > 1. \quad (\text{B10})$$

Nonnegative input barycentric coordinates and output values impose two conditions: the triangle contains both $(1, A)$ and $(-m, B_0)$, and it lies in the wedge

$$A = z, \quad B_0 = z/r_y, \quad m = (7 + 2\rho)/5, \quad (\text{B11})$$

$$L = (3 - \rho/2)/8, \quad R_w = 35\rho/88, \quad (\text{B12})$$

$$v \geq L(-m - u), \quad v \geq R_w(u - 1). \quad (\text{B13})$$

3. Vertex signs, boundary cases and the reflected inequality

A wedge point with nonpositive height has abscissa in $[-m, 1]$. If the triangle had only one positive-height vertex with abscissa u_0 , positive height at input abscissa 1 would require $u_0 \geq 1$: otherwise every convex combination using that vertex with positive weight has abscissa less than one. The other positive input similarly requires $u_0 \leq -m$, a contradiction. The interior origin requires a negative-height vertex. Thus the three vertex heights consist of two strictly positive values and one strictly negative value; zero-height vertices are excluded. The negative vertex is strictly inside the abscissa interval, and the two positive vertices must extend to either end. Consequently they can be labeled

$$(-\ell, P), \quad (c, Q_v), \quad (d, -W), \quad \ell \geq m, \quad c \geq 1, \quad -m < d < 1, \quad P, Q_v, W > 0. \quad (\text{B14})$$

At this stage $\ell = m$, $c = 1$, and $d = 0$ have not been discarded. Put $a = k - 1 > 0$. Wedge membership gives

$$\begin{aligned} P &\geq L(\ell - m), & Q_v &\geq R_w(c - 1), \\ W &\leq \min\{L(m + d), R_w(1 - d)\} < 1. \end{aligned} \quad (\text{B15})$$

The last strict inequality follows from $W < L(m + 1) < (1/4)(7/2) = 7/8$. Lower-face velocities at the two upper vertices give

$$(a\ell + kd)P \leq \ell W, \quad (ac - kd)Q_v \leq cW. \quad (\text{B16})$$

Input containment above and below the relevant edges gives

$$B_0 \leq H_c := \frac{(c+m)P + (\ell-m)Q_v}{\ell+c}, \quad (\text{B17})$$

$$A \leq \frac{(c-1)P + (\ell+1)Q_v}{\ell+c}, \quad (\text{B18})$$

$$(Q_v + W)(1-d) \leq (A+W)(c-d), \quad (\text{B19})$$

$$(P+W)(d+m) \leq (B_0+W)(\ell+d). \quad (\text{B20})$$

All geometric denominators here are strictly positive.

If $d < 0$, the second inequality (B16) gives $Q_v < W/a$. Using explicitly the right-output inequality in (B15) and then (B20), one obtains

$$\begin{aligned} A &\leq Q_v + \frac{(c-1)P}{\ell+c} \leq Q_v \left(1 + \frac{P}{R_w(\ell+c)}\right) \\ &< \frac{W}{a} \left(1 + \frac{B_0+W}{R_w(m+d)}\right) \leq \frac{L}{a} \left(m+d + \frac{B_0+W}{R_w}\right) \\ &< \frac{L}{a} \left(m + \frac{B_0+1}{R_w}\right) < \frac{25/8 + (25/18)(z+1)}{z-7}. \end{aligned} \quad (\text{B21})$$

In the middle step, $(P+W)/(\ell+c) < (B_0+W)/(m+d)$ follows from $\ell+d < \ell+c$. The final comparison uses $m < 5/2$, $L < 1/4$, $R_w > 9/10$, $B_0 < z$ and $a > (z-7)/5$. The last fraction is below $z = A$ for $z \geq 10$, since clearing the positive denominator gives

$$72(z-10)^2 + 836(z-10) + 835 > 0. \quad (\text{B22})$$

This contradiction excludes $d < 0$.

For $d \geq 0$, $a > 7/10$ and (B16) imply $P \leq W/a < 10/7 < 2 < B_0 < A$. Upper containment gives $Q_v \geq A$ and $\ell > m$: otherwise $B_0 \leq H_c = P$. Since $aQ_v - W > 0$, the equivalent inequality $c(aQ_v - W) \leq kdQ_v$ also excludes $d = 0$. Write $\epsilon = 1-d > 0$. From $c \geq 1$ and the same inequality,

$$\epsilon \leq \frac{A+W}{kA} \leq \frac{A+R_w\epsilon}{kA}, \quad \epsilon \leq \frac{1}{k-R_w/A}. \quad (\text{B23})$$

The denominator is positive. The wedge apex satisfies

$$d_0 = \frac{R_w - Lm}{L + R_w} > 0, \quad 1 - d_0 = \frac{L(m+1)}{L + R_w}. \quad (\text{B24})$$

At $z = 21/2$ the latter expression exceeds the last bound in (B23) by more than 0.014259, as certified by rational evaluation in $\mathbb{Q}(\sqrt{6})$. Since $k - R_w/z$ increases with z , every candidate has $d > d_0$. The tighter bottom bound is therefore $W \leq R_w\epsilon$, with equality allowed.

4. The global envelope, including both sides of its pole

Fix d, W from a feasible triangle. The left bounds imply

$$L(\ell - m) \leq g(\ell) := \frac{\ell W}{a\ell + kd}. \quad (\text{B25})$$

Here $g' > 0$, $g'' < 0$, and the left-intersection equation for $\ell_* = m + t$ is a quadratic in t with positive leading coefficient and negative constant. It has precisely one positive root. Thus

$$\ell \leq \ell_*, \quad P \leq P_* := Lt = g(m + t), \quad Lt[a(m + t) + kd] = (m + t)W. \quad (\text{B26})$$

In particular $P_* < W/a < 2$. The partial derivatives

$$\frac{\partial H_c}{\partial P} = \frac{c + m}{\ell + c} > 0, \quad \frac{\partial H_c}{\partial \ell} = \frac{(c + m)(Q_v - P)}{(\ell + c)^2} > 0 \quad (\text{B27})$$

remain strict when P is increased to P_* and ℓ to ℓ_* , because $Q_v \geq A > 2$. This replacement is an upper-bound relaxation; it need not retain every unused constraint. Equality in this relaxation forces both $P = P_*$ and $\ell = \ell_*$.

For the right vertex set

$$\mathcal{L}(c) = \frac{A + W}{\epsilon}(c - d) - W, \quad \mathcal{H}(c) = \frac{cW}{ac - kd}. \quad (\text{B28})$$

The line bounds Q_v for every $c \geq 1$. When $c \leq kd/a$, the invariance inequality has nonpositive left coefficient and imposes no additional bound; one must not divide by it. When $c > kd/a$, it gives $Q_v \leq \mathcal{H}(c)$. The relevant crossover is

$$c_* = \frac{k}{a} \left[d + \frac{W\epsilon}{A + W} \right], \quad Q_*^v = \frac{Ad + W}{a\epsilon}. \quad (\text{B29})$$

Indeed $c_* - kd/a = kW\epsilon/[a(A + W)] > 0$, and on the positive-denominator branch the exact factorization is

$$\mathcal{L}(c) - \mathcal{H}(c) = \frac{a(A + W)(c - d)(c - c_*)}{\epsilon(ac - kd)}. \quad (\text{B30})$$

The other algebraic intersection $c = d$ lies outside $c \geq 1$. Therefore the active bound is the line up to c_* and the hyperbola afterward, including the possible interval below the pole. If $c_* < 1$, the hyperbola at $c = 1$ is less than $\mathcal{L}(1) = A$, and decreases for larger c , contradicting $Q_v \geq A$. Hence $c_* \geq 1$ and $Q_*^v \geq A$.

Write $t = \ell_* - m > 0$. At fixed left maximum,

$$H_c = P_* + \frac{t(Q_v - P_*)}{\ell_* + c}, \quad \partial_{Q_v} H_c = \frac{t}{\ell_* + c} > 0. \quad (\text{B31})$$

Along the line its derivative with respect to c is

$$\frac{t[(A+W)(\ell_*+d)/\epsilon+W+P_*]}{(\ell_*+c)^2} > 0. \quad (\text{B32})$$

Along the hyperbola,

$$\frac{dH_c}{dc} = \frac{t(P_* - \mathcal{H}(c))}{(\ell_*+c)^2} - \frac{tWkd}{(\ell_*+c)(ac-kd)^2} < 0, \quad (\text{B33})$$

because $\mathcal{H}(c) > W/a > P_*$. Thus the unique right maximum is (c_*, Q_*^v) , and equality forces both coordinates to have those values.

It remains to increase W at fixed d . Put $r_c = c_*/Q_*^v = k\epsilon/(A+W)$; the resulting envelope is

$$H_* = \frac{t[Lm + (1 + Lr_c)Q_*^v]}{m + t + r_c Q_*^v}. \quad (\text{B34})$$

Implicit differentiation of (B26) gives $dt/dW > 0$, since the derivative of its left-minus-right expression with respect to t , after using the equality, is

$$L \left[a(m+t) + \frac{kd m}{m+t} \right] > 0. \quad (\text{B35})$$

Also $dQ_*^v/dW = 1/(a\epsilon) > 0$ and $dr_c/dW < 0$. For $H_*(t, Q, r)$, the three relevant partial derivatives have numerators

$$[Lm + (1 + Lr)Q](m + rQ), \quad t(m + t + Lrt), \quad tQ(Lt - Q), \quad (\text{B36})$$

over the common positive square $(m + t + rQ)^2$. Their signs are positive, positive, and negative. These signs hold throughout increasing W to $R_w\epsilon$, since $Lt < W/a < 2 < A \leq Q_*^v$. Thus the increase is strict, and the final bound is attained only at $W = R_w\epsilon$.

5. Discriminant exclusion and propagation of every equality

After the last saturation, put $J = am + k$ and solve (B26) for ϵ :

$$\epsilon = \frac{Lt(J + at)}{R_w m + (kL + R_w)t}, \quad H_* - B_0 = \frac{f_0 + f_1 t + f_2 t^2}{(J + at)^2}. \quad (\text{B37})$$

All denominators are positive. Direct rational substitution gives

$$f_0 = AR_w m/L - B_0 J^2, \quad (\text{B38})$$

$$f_1 = A(R_w/L + k - am) + LJ^2 + R_w J - 2B_0 J a, \quad (\text{B39})$$

$$f_2 = a[LJ + R_w - A(1 + Lk/R_w) - B_0 a]. \quad (\text{B40})$$

In particular

$$f_0 = -\frac{19+9\rho}{100}z \left(z^2 - 11z + \frac{211}{44} \right), \quad (\text{B41})$$

$$f_1^2 - 4f_0f_2 = -\left(\frac{11}{1200} + \frac{11\rho}{4200} \right) \left(z + \frac{19}{22} \right)^2 \frac{\mathcal{P}(z)}{352}. \quad (\text{B42})$$

Descartes' rule and opposite signs at zero and positive infinity imply that $\mathcal{P}$ has exactly one positive root. Rational endpoint signs isolate it in $(10.55714966866, 10.55714966867)$. The positive root of the quadratic in (B41) is below this interval: that quadratic is positive at the lower endpoint and strictly increasing there and above. Hence for every $z > z_*$, $f_0 < 0$ and the discriminant is negative. The quadratic has no real zero and is negative at zero, so it is negative everywhere. The necessary envelope bound excludes every connected hidden support, without any boundedness assumption on the candidate triangle coordinates.

At $z = z_*$, exact interval evaluation gives

$$-0.520929 < f_0 < -0.520928, \quad 5.496508 < f_1 < 5.496509, \quad -14.498918 < f_2 < -14.498917. \quad (\text{B43})$$

The numerator is nonpositive and vanishes only at $t_* = -f_1/(2f_2) > 0$. Start with any feasible triangle and follow the three upper-bound relaxations above. Its original upper input height is at least B_0 ; its final saturated height is at most B_0 . Writing H_*^{sat} for Eq. (B34) after $W = R_w(1-d)$, the chain is

$$B_0 \leq H_c \leq H_* \leq H_*^{\text{sat}} \leq B_0. \quad (\text{B44})$$

Thus every relaxation is equality. Strict increase under W saturation first forces the original $W = R_w(1-d)$. The final numerator forces $t = t_*$. Equation (B37) now fixes ϵ and hence d, W uniquely. Equality in the left and right relaxations fixes $(\ell, P) = (m+t_*, Lt_*)$ and $(c, Q_v) = (c_*, Q_*^v)$. This explicitly rules out equality lost in any intermediate relaxation.

The left and bottom vertices lie on the left and right output faces. Input $(1, A)$ lies on the opposite lower-right edge, and input $(-m, B_0)$ lies on the opposite upper edge. The paired input/output equalities remove the same two visible reciprocal pairs. The two lower-face tangencies remove both directions between the upper vertices. These are necessary exact zeros; positivity of the remaining rates is an additional existence check, not a consequence of the zeros alone.

Finally a fixed triangle leaves no free positive column scaling. For its sectioned vertex matrix V , set $\bar{H} = V^{-1}DV$, $\bar{Y} = V^{-1}Y_m$, $\bar{X} = X_mV$. If its columns are rescaled by a positive vector w , hidden row normalization requires $\bar{H}w + \bar{Y}\mathbf{1} = 0$. Since all three hidden eigenvalues are nonzero,

TABLE V. Endpoint equality conditions. Every row is necessary for a physical triangle at $z = z_*$; existence is checked separately in Appendix C.

Equality step	Reason and forced parameters
$H_* = H_*^{\text{sat}}$	Strict increase with W at fixed d forces $W = R_w(1 - d)$.
$H_*^{\text{sat}} = B_0$	The quadratic has its sole zero at $t_* = -f_1/(2f_2)$; Eq. (B37) fixes d and W .
$H_c = H_*$	Strict left-coordinate increase forces $(\ell, P) = (m + t_*, Lt_*)$; the unique right maximum forces $(c, Q_v) = (c_*, Q_*^v)$.
$B_0 = H_c$	The upper input lies on the triangle's upper edge.
Physical normalization	Eq. (B45) uniquely fixes the column scaling; hidden permutation remains.

there is exactly one possible scaling:

$$w = -\bar{H}^{-1}\bar{Y}\mathbf{1}, \quad W_d = \text{diag}(w), \quad H' = W_d^{-1}\bar{H}W_d, \quad X' = \bar{X}W_d, \quad Y' = W_d^{-1}\bar{Y}. \quad (\text{B45})$$

Hidden permutation is the only remaining coordinate freedom. Thus the equality triangle gives at most one physical connected-hidden generator. Appendix C supplies its existence, positive scaling, exact reciprocal support and entropy separation independently. Together with the disconnected classification, this proves the endpoint and supercritical conclusions over the entire stated model class.

Appendix C: Exact endpoint construction and entropy certificate

An explicit rational definition avoids using rounded rates. In the following equations evaluate all expressions at $z = z_*$. Define

$$C = \begin{pmatrix} 3 & 0 & 1 \\ 0 & 6 & 1 \\ z & z & 1 \end{pmatrix}, \quad D_c = C^{-1}HC, \quad (\text{C1})$$

where H is (A4). Use one-based indices and set

$$(a_1, a_2, b_1, b_2, c_1, c_2) = ((D_c)_{22}, (D_c)_{21}, (D_c)_{12}, (D_c)_{11}, (D_c)_{32}, (D_c)_{31}). \quad (\text{C2})$$

Let

$$a_p(z) = 100z^4 - 840z^3 + 3741z^2 - 6867z + 3978, \quad (\text{C3})$$

$$b_p(z) = -408z^4 - 2916z^3 + 8766z^2 + 3105z - 12474, \quad (\text{C4})$$

$$c_p(z) = 2412z^4 - 5724z^3 + 891z^2 + 3402z. \quad (\text{C5})$$

They satisfy the independent physical-path discriminant identity

$$b_p^2 - 4a_p c_p = -567(z-2)^2(2z-3)^2 \mathcal{P}(z). \quad (\text{C6})$$

Set

$$u = -\frac{b_p}{2a_p}, \quad v = -\frac{u(a_2 u + c_2)}{u^2 + a_1 u + c_1}, \quad (\text{C7})$$

$$p_x = (42 + 11z)/18, \quad p_y = 11z/18, \quad q_x = 8z/10, \quad q_y = (6 + 8z)/10, \quad (\text{C8})$$

$$c'_p = \frac{p_x - u}{p_y}, \quad d'_p = \frac{q_y - v}{q_x}, \quad A_p = \frac{u + c'_p v}{1 - c'_p d'_p}, \quad B_p = \frac{v + d'_p u}{1 - c'_p d'_p}. \quad (\text{C9})$$

The primed c'_p, d'_p are chart coordinates, not the polynomial $c_p(z)$. Finally define

$$C' = \begin{pmatrix} 0 & u & 1 \\ v & 0 & 1 \\ B_p & A_p & 1 \end{pmatrix}, \quad U = C(C')^{-1}, \quad S = \text{diag}(I_2, U), \quad Q_* = S^{-1}Q(z_*)S. \quad (\text{C10})$$

The hidden order is (a, b, c) : a y -only endpoint, an x -only center, and a shared endpoint. The seven reciprocal edges are $xy, xb, xc, ya, yc, ab, bc$. All other off-diagonals are exactly zero. For orientation,

$$Q_* \simeq \begin{pmatrix} -11 & 1 & 0 & 1.86553087 & 8.13446913 \\ 2 & -20 & 6.92248992 & 0 & 11.07751008 \\ 0 & 6.22824145 & -11.27299011 & 5.04474866 & 0 \\ 2.77737298 & 0 & 3.73491843 & -6.73129745 & 0.21900604 \\ 10.48328058 & 10.38263173 & 0 & 0.24409946 & -21.11001177 \end{pmatrix}. \quad (\text{C11})$$

This rounded matrix is not the equality certificate.

The following is a finite, reproducible algebraic admissibility check. Work in the real field $\mathbb{Q}(z_*)$ with $\mathcal{P}(z_*) = 0$, reducing each element to a polynomial of degree at most three. Isolate the positive root in

$$\left[\frac{6329583361674357233}{599554194108037085}, \frac{5875360061222171594}{556529010729195633} \right]. \quad (\text{C12})$$

The interval has width below 10^{-35} ; opposite polynomial signs and unique positivity select the embedding. Rational interval evaluation verifies that $\det C = 18 - 9z$, a_p , $u^2 + a_1 u + c_1$, p_y, q_x , $1 - c'_p d'_p$, and $\det C'$ are all nonzero. Reduce (C10) exactly: $U\mathbf{1} = \mathbf{1}$, row sums vanish, the six absent off-diagonal entries are zero, and the fourteen remaining off-diagonals are positive. Their enclosures lie strictly between $1/5$ and 12 . The stated seven-edge graph is connected. This

suffices for admissibility. The package supplies every reduced coefficient, making these finite checks repeatable without solving a new realization problem.

Compute the unique normalized stationary vectors by exact reduced linear solves. Both are positive and satisfy $\pi_* = \pi S$. Directly verify $RT^j B = RT_*^j B$ for $j = 0, \dots, 9$; Lemma 14 gives all-time equality independently of the displayed similarity. As a separate coordinate check, the equality triangle in Appendix B, normalized by (B45), agrees coefficientwise with (C10) after swapping its last two hidden vertices. This computation uses pairs $a + b\sqrt{6}$ over $\mathbb{Q}(z_*)$; every final $\sqrt{6}$ coefficient cancels.

Entropy separation also needs a certificate, since different generators need not have different entropy. Evaluate stationary currents and positive flux ratios with rational intervals from (C12). To bound a logarithm, reduce its positive rational endpoints by powers of two into $[1, 2]$. For $t = (x - 1)/(x + 1) \in [0, 1/3]$ use

$$\log x = 2 \sum_{j=0}^{n-1} \frac{t^{2j+1}}{2j+1} + R_n, \quad 0 \leq R_n \leq \frac{2t^{2n+1}}{(2n+1)(1-t^2)}, \quad n = 24. \quad (\text{C13})$$

Bound $\log 2$ by the same formula. Interval arithmetic accounts for the signs of powers-of-two shifts, currents, and products, and sums once per unordered present edge. It gives (64)–(65) and

$$0.002633285729 \leq \sigma(Q_*) - \sigma(Q(z_*)) \leq 0.002633285730. \quad (\text{C14})$$

These are rational enclosures, not precision-based equality tests. Together with Appendix B, they finish the endpoint and supercritical parts of Theorem 11.

Appendix D: An exact entropy-divergent hidden-pair orbit

Take fully hidden states h, k with the same external neighbor set J , and different full external exit rows Y_h, Y_k . Let $m_0 = q_{hk}$, $n_0 = q_{kh}$, and let

$$\lambda_h = m_0 + \sum_{i \in J} Y_{hi}, \quad \lambda_k = n_0 + \sum_{i \in J} Y_{ki} \quad (\text{D1})$$

be their *total* escape rates. Assume either $m_0, n_0 > 0$, or $m_0 = n_0 = 0$ with $\lambda_h = \lambda_k$. The external rows include all other hidden states, not only visible endpoints.

Exchange h, k if necessary so that $r_* := \min_{i \in J} Y_{hi}/Y_{ki} \in (0, 1)$. Such an orientation exists because the rows differ and their supported entries are positive. Embed

$$U_e = \begin{pmatrix} 1-e & e \\ 0 & 1 \end{pmatrix}, \quad 0 \leq e < r_*, \quad (\text{D2})$$

as the hidden block of S_e . The transformed outside entrances and exits are

$$q'_{ih} = (1 - e)q_{ih}, \quad q'_{ik} = q_{ik} + eq_{ih}, \quad (\text{D3})$$

$$q'_{hi} = (Y_{hi} - eY_{ki})/(1 - e), \quad q'_{ki} = Y_{ki}. \quad (\text{D4})$$

They retain their reciprocal support before r_* . For an adjacent pair,

$$q'_{kh} = n_0(1 - e), \quad q'_{hk} = \frac{f(e)}{1 - e}, \quad f(e) = m_0 + e(\lambda_k - \lambda_h) - n_0e^2. \quad (\text{D5})$$

Let e_* be the first zero of f in $(0, r_*]$, if present, and otherwise r_* . Then $0 < e_* \leq r_* < 1$, and all generators before e_* retain the original support. For the unjoined equal-escape pair the hidden block is scalar and unchanged; take $e_* = r_*$.

Normalized similarity preserves the exact kernel. The stationary law is explicitly

$$\pi'_i = \pi_i \ (i \notin \{h, k\}), \quad \pi'_h = (1 - e)\pi_h, \quad \pi'_k = \pi_k + e\pi_h. \quad (\text{D6})$$

Every stationary mass has a positive finite limit and all rates remain bounded because $e_* < 1$. If $f(e_*) = 0$, the $h \rightarrow k$ flux vanishes while its reverse tends to $(\pi_k + e_*\pi_h)n_0(1 - e_*) > 0$. If $e_* = r_*$, an external exit $h \rightarrow i$ vanishes while its reverse tends to $\pi_i(1 - r_*)q_{ih} > 0$. This also covers simultaneous zeros. The corresponding nonnegative entropy term diverges, and other terms cannot cancel it. The endpoint itself need not be admitted or have a unique stationary law; (D6) gives the necessary limits before the boundary.

For a minimal realization, equal full external exit rows are impossible: the codimension-one subspace of vectors with equal h, k entries is invariant under T and contains every column of B . Reachability would fail. Thus every pair in a complete minimal realization meets the unequal-row requirement. This observation is used below, while Lemma 6 creates unequal rows explicitly and does not assume minimality.

An explicit complete-support example separates exact ambiguity from the three-state instability under observation error. For $0 \leq e < 2/3$, let

$$Q_e = \begin{pmatrix} -6 & 1 & 2(1 - e) & 3 + 2e \\ 2 & -9 & 3(1 - e) & 4 + 3e \\ (4 - 6e)/(1 - e) & 5 & -15 - 4e & (6 - 4e^2)/(1 - e) \\ 6 & 5 & 4(1 - e) & -15 + 4e \end{pmatrix}. \quad (\text{D7})$$

All twelve off-diagonals are positive, bounded, and row-normalized. Observing only $x \leftrightarrow y$ in state order (x, y, h, k) gives $\Psi_{Q_e} = \Psi_{Q_0}$ for every such e , by the preceding shear or by direct equality

of the full resolvents. Only q_{hx} tends to zero; $q_{xh} \rightarrow 2/3$ and $\pi_x = 557/1420$. The reverse flux vanishes linearly in $2/3 - e$, while the forward flux tends to $557/2130 > 0$. All other edge-flux ratios have positive finite limits. Consequently

$$\sigma(Q_e) = \frac{557}{2130} \log \frac{1}{2/3 - e} + O(1). \quad (\text{D8})$$

This is an exact compatible path within complete support, approaching an inadmissible one-way boundary. Its full kernel identity and stationary flux limit are independently checked from the displayed generator; 160-digit entropy evaluations are a further numerical check of the limit.

Appendix E: Complete subcritical coverage, including repeated poles

We give four covering constructions. Each preserves the hidden transfer exactly and yields a pair to which Appendix D applies.

1. Below the slow pole

Take $0 < z < \alpha/2$. Let $d_s = 2(\rho - 1)$. In hidden order (h, k, l) set

$$K_{hl} = -4, \quad K_{kl} = -d_s, \quad K_{lh} = 5, \quad K_{lk} = d_s, \quad K_{hk} = K_{kh} = 0, \quad (\text{E1})$$

and fix diagonals by $K\mathbf{1} = 0$. Under $U = I + \epsilon K$, the derivative of the transformed generator is the commutator with embedded K . The four missing hidden rates have derivatives $(\alpha - 2z)(4, d_s, 5, d_s) > 0$; the missing visible incidences (xk, yh, hy, kx) have derivatives $(8d_s, 55, 4z, d_s z) > 0$. For sufficiently small $\epsilon > 0$, U is invertible, all missing pairs open, and existing positive rates remain positive. The target is complete and minimal, hence has an entropy-divergent hidden pair. This interval includes the path-chart singularity $z = 2$.

2. Between the hidden poles

Take $\alpha/2 \leq z \leq \beta/2$. Put $s_0 = 2z$, $d_s = 2(\rho - 1)$, and $\Delta = 4\rho$. Factor the two relevant residues as

$$Z_{s_0} = \begin{pmatrix} 8 \\ 11 \end{pmatrix} (z, z), \quad Z_\beta = \begin{pmatrix} 2d_s \\ -35 \end{pmatrix} \frac{(3, -24/d_s)}{\Delta}. \quad (\text{E2})$$

Choose midpoints in the nonempty intervals

$$\frac{24}{d_s \Delta z} < t < \frac{4}{d_s}, \quad \frac{3}{\Delta z} < u < \frac{11}{35}. \quad (\text{E3})$$

They are nonempty because $z > \rho/4$ and $z > 35\rho/88$, both already true at $\alpha/2$. For $P_2 = \begin{pmatrix} 1 & 1 \\ -t & u \end{pmatrix}$, use

$$\bar{X} = \begin{pmatrix} 8 - 2d_s t & 8 + 2d_s u \\ 11 + 35t & 11 - 35u \end{pmatrix} > 0, \quad (\text{E4})$$

$$\bar{Y} = \frac{1}{t+u} \begin{pmatrix} uz - 3/\Delta & uz + 24/(d_s \Delta) \\ tz + 3/\Delta & tz - 24/(d_s \Delta) \end{pmatrix} > 0, \quad (\text{E5})$$

$$\bar{H} = \frac{1}{t+u} \begin{pmatrix} -s_0 u - \beta t & u(\beta - s_0) \\ t(\beta - s_0) & -s_0 t - \beta u \end{pmatrix}. \quad (\text{E6})$$

These are obtained by similarity from modes $-s_0, -\beta$. The stable Metzler $\bar{H}$ has two positive off-diagonals for $s_0 < \beta$, and is $-\beta I$ at equality. Normalize by (B45), giving a pair with strictly positive visible incidences. Its exit matrix has rank two because the modal exit rows are a positive row and a mixed-sign row; their independence is preserved.

For the remaining residue write

$$Z_\alpha = cg, \quad c = \begin{pmatrix} 2(d_s + 4) \\ 35 \end{pmatrix}, \quad g = (3, 24/(d_s + 4))/\Delta. \quad (\text{E7})$$

A singleton with entrance $(g\mathbf{1})c/\alpha$ and exit $\alpha g/(g\mathbf{1})$ realizes it. Adjoin it without hidden links to the pair, retaining the source visible block. Hidden row normalization and equality of the transfer at zero give the source's visible hidden-entry totals $(10, 18)^T$, so visible rows also normalize. The full generator is irreducible through its visible links. The pair has unequal external exits, is adjacent for $s_0 < \beta$, and unjoined with equal escapes at $s_0 = \beta$. Appendix D applies in both cases. All transfer identities remain valid when poles coincide: the residues simply add. Thus both endpoints, including $s_0 = \alpha$, are covered without a distinct-pole assumption.

3. An explicit fast opening

Take $\beta/2 \leq z \leq 21/2$. Put $M_s = 2z - 11 > 0$, $L_s = M_s + 4$, and define

$$v_{\max} = 7z/33, \quad q = \frac{1}{2}(M_s/5 + 96v_{\max}/(zM_s)), \quad v = \frac{1}{2}(zqM_s/96 + v_{\max}), \quad (\text{E8})$$

$$u = \frac{1}{2}(zq/24 + 4v/M_s), \quad p = \frac{1}{2}(-4u - zq/6), \quad r = \frac{1}{2}(11v/7 + z/3). \quad (\text{E9})$$

Use hidden off-diagonals $(K_{hk}, K_{hl}, K_{kh}, K_{kl}, K_{lh}, K_{lk}) = (p, q, r, -1, -v, u)$ and $K\mathbf{1} = 0$. Since $L_s M_s \geq 20$ and $11M_s^2 \leq 1100 < 1120$, the midpoint intervals are strict:

$$q > M_s/5, \quad v < 7z/33, \quad zq/24 < u < 4v/M_s, \quad (\text{E10})$$

$$-4u < p < -zq/6, \quad 11v/7 < r < z/3. \quad (\text{E11})$$

The eight missing-rate derivatives, ordered $(xk, yh, hy, hl, kx, kl, lh, lk)$, are

$$(2p + 8u, 7r - 11v, -6p - zq, L_s q - 4, -3r + z, 5q - M_s, L_s v - 5u, 4v - M_s u), \quad (\text{E12})$$

all positive. Thus $I + \epsilon K$ opens a complete realization for small $\epsilon > 0$, with exact observation and subsequent entropy divergence as above. This argument needs neither an exit chart nor an assumption that failure of a first-order opening implies global rigidity.

4. Strict triangles up to the fold

Take $21/2 \leq z < z_*$. Use the quantities of Appendix B and set $t = -f_1/(2f_2)$. On the entire rational interval

$$I = [10.5, 10.55714966867], \quad (\text{E13})$$

rational interval evaluation gives $f_1 > 5$ and $f_2 < -14$. For $z < z_*$, the positive discriminant in (B42) gives

$$f_0 + f_1 t + f_2 t^2 = -\frac{f_1^2 - 4f_0 f_2}{4f_2} > 0. \quad (\text{E14})$$

Define $\epsilon, d, W, \ell, P, c, Q_v$ by (B37), $d = 1 - \epsilon$, $W = R_w \epsilon$, $\ell = m + t$, $P = Lt$, and (B29). Then $B_0 < H_*$ is strict upper-input slack. The remaining geometric signs hold throughout I ; Table VI gives conservative rational lower bounds. To reproduce them, enclose ρ between the exact decimal rationals 2.449489742783178098197284 and 2.449489742783178098197285, whose squares straddle six. Shift polynomial coefficients by $z = 21/2 + \xi$, and apply rational interval operations, rounded outward to 24 decimal places after each operation, with $0 \leq \xi \leq 0.05714966867$. No parameter sampling establishes these bounds.

For completeness, the left and right lower-face tangencies imply $p_\ell \ell = kP$ and $q_r c = kQ_v$. Therefore the lower-face margins of the origin are $W - p_\ell d = aP > 0$ and $W + q_r d = aQ_v > 0$. The upper edge is above the origin, so the triangle is nondegenerate with an interior origin. The inward derivatives at the bottom vertex are $a(W + P) > 0$ and $a(W + Q_v) > 0$. The upper-edge inward derivatives at its endpoints have positive numerators $[a\ell + kc]P + \ell Q_v$ and $[k\ell + ac]Q_v + cP$. The two remaining derivatives vanish by tangency. These vertex-face conditions suffice for invariance under the linear flow.

The initial triangle is nonnegative but can have one-way support. Make three small changes for each fixed $z < z_*$. First decrease W while holding d fixed, recomputing (B26) and (B29); all

TABLE VI. Strict rational lower bounds valid on the whole interval (E13), with $p_\ell = (P + W)/(\ell + d)$, $q_r = (Q_v + W)/(c - d)$. The upper-input- B_0 slack is not listed because it vanishes at the fold.

Quantity	Lower bound	Quantity	Lower bound
t	18/100	ϵ	5/100
$1 - \epsilon$	9/10	$c - 1$	1
$L(m + d) - W$	6/10	$Q_v - R_w(c - 1)$	170
$B_0 + W - p_\ell(d + m)$	8	$[(c - 1)P + (\ell + 1)Q_v]/(\ell + c) - A$	100
$W - p_\ell d$	3/100	$W + q_r d$	120
$kW - p_\ell d$	7/100	$kW + q_r d$	120

strict margins persist and the bottom output becomes strict. Next decrease ℓ slightly and take P to be the midpoint of $L(\ell - m)$ and $\ell W/(a\ell + kd)$; both left output and left inward inequalities become strict. Finally keep $c = c_*$ and decrease Q_v slightly; since $ac - kd > 0$, both the right inward inequality and the lower containment of $(1, A)$ become strict. All other strict inequalities persist by continuity. Only finitely many conditions are involved, so successive sufficiently small positive choices exist; no uniform size as $z \uparrow z_*$ is claimed.

The final triangle has strictly positive input barycentric coordinates, output values, and all off-diagonal inward derivatives. At vertex j , the derivative of each initially zero barycentric coordinate $i \neq j$ is $(\bar{H})_{ij}$ up to the positive time-rescaling factor; thus strict inward derivatives give positive off-diagonals. Consequently $\bar{X} > 0$, $\bar{Y} > 0$, and every off-diagonal of the stable $\bar{H}$ is positive. The normalization vector in (B45) is positive because $-\bar{H}^{-1} > 0$. Hidden row sums vanish and transfer equality at zero fixes the visible row sums. This is a complete five-state physical generator with the same all-time observation. It is minimal by Appendix A, so its hidden pair meets Appendix D. The four intervals cover all $0 < z < z_*$, completing Theorem 11.

Appendix F: Why one additional state changes the exact answer

Let h be fully hidden in any irreducible generator. Replace it by h_1, h_2 ; keep rates among outside states, split each incoming rate as $\theta_i q_{ih}$ and $(1 - \theta_i)q_{ih}$ with $0 < \theta_i < 1$, and give both clones the original outgoing rates q_{hj} to every outside j . Set internal rates $q_{h_1 h_2} = m > 0$ and $q_{h_2 h_1} = n > 0$, with n fixed. Normalize all rows. Every positive m gives connected reciprocal support with the same observed edges.

For the aggregation matrix C merging the clones, $\tilde{Q}C = CQ$. Because no mark touches h , also

$\tilde{T}C = CT$, $\tilde{B} = CB$, and $\tilde{R}C = R$. Therefore $\tilde{\Psi}(t) = \tilde{R}e^{\tilde{T}t}CB = Re^{Tt}B$ exactly. This is strong lumping applied to the killed generator, not just agreement of stationary probabilities.

Put $\lambda = \sum_{j \neq h} q_{hj}$ and $\alpha_c = \sum_{i \neq h} \pi_i \theta_i q_{ih} / \pi_h \in (0, \lambda)$. Outside stationary probabilities remain π_i , and clone probabilities are

$$\tilde{\pi}_{h_1} = \pi_h \frac{\alpha_c + n}{\lambda + m + n}, \quad \tilde{\pi}_{h_2} = \pi_h \frac{\lambda + m - \alpha_c}{\lambda + m + n}. \quad (\text{F1})$$

Both have positive limits as $m \downarrow 0$. The first internal flux tends to zero and the reverse tends to $\pi_h n(\lambda - \alpha_c)/(\lambda + n) > 0$, so entropy diverges while rates remain uniformly bounded above. Applied to $Q(z_*)$, this proves unboundedness with six states. These enlarged realizations are nonminimal for the common marked kernel. This appendix asserts sufficiency of one extra state, not a general necessary state budget.

Appendix G: Local persistence of the five-state boundary

Proof. In Appendix B replace only A and B_0 by

$$A = z + \delta, \quad B_0 = (z - \delta)/r_y. \quad (\text{G1})$$

The other modal constants, the flow, and the wedge are unchanged. Minimality persists: one hidden reachability minor and the observability minor are

$$\det(Y_1, Y_2, HY_1) = -9(4\delta z - 19\delta + 4z^2 - 9z), \quad (\text{G2})$$

$$\det \begin{pmatrix} X_{1,:} \\ X_{2,:} \\ (XH)_{2,:} \end{pmatrix} = -14(22z + 19). \quad (\text{G3})$$

They remain nonzero near the critical point. The disconnected-support argument is unchanged: the two fixed slow residues determine their pair, and the fast singleton has exits $(z + \delta, z - \delta)$.

The envelope coefficients (B40) were derived for independent input heights. Their discriminant now obeys the exact identity

$$\Delta(z, \delta) := f_1^2 - 4f_0f_2 = - \left(\frac{11}{1200} + \frac{11\sqrt{6}}{4200} \right) \left(z + \frac{19}{22} \right)^2 \frac{\mathcal{P}_\delta(z)}{352}. \quad (\text{G4})$$

For completeness, the accompanying rational interval certificate verifies the strict reductions and construction margins on the entire rectangle

$$\mathcal{R} = [10.55, 10.565] \times [-0.005, 0.005]. \quad (\text{G5})$$

In particular $A > B_0 > 2$, $a > 7/10$,

$$A - \frac{L}{a} \left[m + \frac{B_0 + 1}{R_w} \right] > 6.691, \quad \frac{L(m+1)}{L + R_w} - \frac{1}{k - R_w/A} > 0.0182. \quad (\text{G6})$$

These are uniform reductions for every candidate triangle: they exclude negative bottom abscissas and place the bottom to the right of the wedge apex. The same envelope maximizations therefore apply to all connected supports. The certificate also gives $-f_0 > 0.102$, $f_1 > 5.267$, $-f_2 > 14.445$, and strictly positive versions of every nonactive margin listed in Table VI. Its exact rational endpoints, rather than these shortened decimal summaries, establish the signs.

Furthermore $\partial_z \mathcal{P}_\delta > 486081$ on $\mathcal{R}$, and $\mathcal{P}_\delta$ has opposite signs at its two z endpoints uniformly in δ . Thus each $|\delta| \leq 0.005$ has one critical root in the rectangle. Nonzero $\partial_z \mathcal{P}_\delta$ gives a smooth root curve. At $\delta = 0$ this also follows from

$$\mathcal{P}'(10 + x) = 1408x^3 + 32736x^2 + 223032x + 353380 > 0 \quad (x \geq 0), \quad (\text{G7})$$

which proves (69) without a floating-point rank test.

Above the curve $\Delta < 0$, so the envelope is negative everywhere and all connected supports are excluded. At the curve its sole zero is $t_c = -f_1/(2f_2) > 0$. Equality fixes one triangle and its unique positive normalization. The active input/output and tangent-face equalities delete the same three reciprocal pairs as in the balanced endpoint; the strictly positive remaining margins prove existence and positivity of all remaining rates. Below the curve, $\Delta > 0$ and $F(t_c) = -\Delta/(4f_2) > 0$. The three successive vertex changes in Appendix E again give a complete positive realization. Its exact-law hidden-pair shear makes entropy unbounded.

This proves the corresponding generator statements on $\mathcal{R}$. At the critical curve the source and the path have fixed reciprocal supports and strictly positive present rates, so their stationary laws and entropy rates depend continuously on δ . The strictly positive certified entropy gap at $\delta = 0$ therefore remains positive in a possibly smaller neighborhood. This establishes the stated two-value entropy assertion; the geometry certificate alone is not an interval certificate for that entropy gap on all of $\mathcal{R}$. $\square$

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- [1] J. van der Meer, B. Ertel, and U. Seifert, Thermodynamic inference in partially accessible Markov networks: A unifying perspective from transition-based waiting time distributions, *Phys. Rev. X* **12**, 031025 (2022), [arXiv:2203.12020](https://arxiv.org/abs/2203.12020).

- [2] P. E. Harunari, A. Dutta, M. Poletti, and É. Roldán, What to learn from a few visible transitions' statistics?, *Phys. Rev. X* **12**, 041026 (2022), [arXiv:2203.07427](#).
- [3] A. M. Maier, U. Seifert, and J. van der Meer, From observed transitions to hidden paths in Markov networks, *Phys. Rev. Research* **7**, 033067 (2025), [arXiv:2504.16015](#).
- [4] M. Wagner and J. Timmer, The effects of non-identifiability on testing for detailed balance in aggregated Markov models for ion-channel gating, *Biophys. J.* **79**, 2918 (2000).
- [5] J. Ehrich, Tightest bound on hidden entropy production from partially observed dynamics, *J. Stat. Mech.: Theory Exp.* **2021** (8), 083214, [arXiv:2105.08803](#).
- [6] E. Nitzan, A. Ghosal, and G. Bisker, Universal bounds on entropy production inferred from observed statistics, *Phys. Rev. Research* **5**, 043251 (2023), [arXiv:2212.01783](#).
- [7] B. Vanluyten, J. C. Willems, and B. De Moor, Equivalence of state representations for hidden Markov models, *Syst. Control Lett.* **57**, 410 (2008).
- [8] M. Telek and G. Horváth, A minimal representation of Markov arrival processes and a moments matching method, *Perform. Eval.* **64**, 1153 (2007).
- [9] Q.-M. He and H. Zhang, PH-invariant polytopes and Coxian representations of phase type distributions, *Stoch. Models* **22**, 383 (2006).
- [10] D. Mond, J. Smith, and D. van Straten, Stochastic factorizations, sandwiched simplices and the topology of the space of explanations, *Proc. R. Soc. Lond. A* **459**, 2821 (2003).
- [11] S. Zeraati, F. H. Jafarpour, and H. Hinrichsen, Entropy production of nonequilibrium steady states with irreversible transitions, *J. Stat. Mech.: Theory Exp.* **2012** (12), L12001, [arXiv:1211.4701](#).
- [12] I. Sason and S. Verdú, f -divergence inequalities, *IEEE Trans. Inf. Theory* **62**, 5973 (2016), [arXiv:1508.00335](#).
- [13] A. C. Barato and U. Seifert, Thermodynamic uncertainty relation for biomolecular processes, *Phys. Rev. Lett.* **114**, 158101 (2015), [arXiv:1502.05944](#).
- [14] T. R. Gingrich, J. M. Horowitz, N. Perunov, and J. L. England, Dissipation bounds all steady-state current fluctuations, *Phys. Rev. Lett.* **116**, 120601 (2016), [arXiv:1512.02212](#).
- [15] I. A. Martínez, G. Bisker, J. M. Horowitz, and J. M. R. Parrondo, Inferring broken detailed balance in the absence of observable currents, *Nat. Commun.* **10**, 3542 (2019), [arXiv:1809.02084](#).
- [16] D. J. Skinner and J. Dunkel, Improved bounds on entropy production in living systems, *Proc. Natl. Acad. Sci. U.S.A.* **118**, e2024300118 (2021), [arXiv:2011.08765](#).
- [17] D. J. Skinner and J. Dunkel, Estimating entropy production from waiting time distributions, *Phys. Rev. Lett.* **127**, 198101 (2021), [arXiv:2105.08681](#).
- [18] J. H. Fritz, B. Ertel, and U. Seifert, Entropy estimation for partially accessible Markov networks based on imperfect observations: Role of finite resolution and finite statistics, *Phys. Rev. E* **111**, 044106 (2025), [arXiv:2412.04102](#).
- [19] Q. Yu and P. E. Harunari, Dissipation at limited resolutions: Power law and detection of hidden dissipative scales, *J. Stat. Mech.: Theory Exp.* **2024**, 103201, [arXiv:2407.13707](#).

- [20] E. Gassiat, A. Cleyne, and S. Robin, Inference in finite state space non parametric hidden Markov models and applications, *Stat. Comput.* **26**, 61 (2016).
- [21] K. Abraham, E. Gassiat, and Z. Naulet, Fundamental limits for learning hidden Markov model parameters, *IEEE Trans. Inf. Theory* **69**, 1777 (2023), [arXiv:2106.12936](#).
- [22] W. J. Bruno, J. Yang, and J. E. Pearson, Using independent open-to-closed transitions to simplify aggregated Markov models of ion channel gating kinetics, *Proc. Natl. Acad. Sci. U.S.A.* **102**, 6326 (2005).
- [23] Q.-M. He and H. Zhang, An algorithm for computing minimal Coxian representations, *INFORMS J. Comput.* **20**, 179 (2008).
- [24] K. Blom, K. Song, E. Vouga, A. Godec, and D. E. Makarov, Milestoning estimators of dissipation in systems observed at a coarse resolution, *Proc. Natl. Acad. Sci. U.S.A.* **121**, e2318333121 (2024), [arXiv:2310.06833](#).
- [25] G. Bisker, I. A. Martínez, J. M. Horowitz, and J. M. R. Parrondo, Reply to: Comment on “inferring broken detailed balance in the absence of observable currents”, *Nat. Commun.* **15**, 8679 (2024).
- [26] J. H. Fritz and U. Seifert, Waiting-time-based entropy estimators in continuous space without Markovian events, *Phys. Rev. E* **114**, 034109 (2026), [arXiv:2603.05170](#).
- [27] K. Kawaguchi and Y. Nakayama, Fluctuation theorem for hidden entropy production, *Phys. Rev. E* **88**, 022147 (2013), [arXiv:1209.6333](#).
- [28] J. Schnakenberg, Network theory of microscopic and macroscopic behavior of master equation systems, *Rev. Mod. Phys.* **48**, 571 (1976).
- [29] T. Nishiyama and Y. Hasegawa, Upper bound for entropy production in Markov processes, *Phys. Rev. E* **108**, 044139 (2023), [arXiv:2306.15251](#).
- [30] J. Moss, Infinite diameter confidence sets in Hedges’ publication bias model, *J. Korean Stat. Soc.* **51**, 946 (2022).
- [31] M. Esposito, Stochastic thermodynamics under coarse graining, *Phys. Rev. E* **85**, 041125 (2012), [arXiv:1112.5410](#).